

HIV and AIDS related knowledge, attitudes and behaviours of students and staff at South African Technical and Vocational Education and Training colleges in South Africa, 2014



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Statement by the Deputy Minister of Higher Education, Mduduzi Manana

The first-ever study on HIV and AIDS-related knowledge, attitudes and behaviours of students and staff at South Africa's technical and vocational education and training colleges (TVETs) was undertaken by the HEAIDS programme, which is the initiative of the Department of Higher Education and Training. In 2008-9, HEAIDS conducted a similar knowledge, attitudes, behaviours and perceptions survey in the universities sub-sector, and hence this study deepens our understanding of the higher education and training sector's health and wellness needs and priorities. The research signals the commitment of the department to address the complexities of the post-schooling sector and deliver on its mandate, which is to produce high-quality graduates and future leaders for society.



We are working across the higher education sector to mainstream health and wellness education and services. Our mission in this regard is to reduce the impact of HIV/AIDS, TB, STIs, and concomitant issues of drug and alcohol abuse, gender violence and stigma towards minority populations. This will help us to achieve desired educational outcomes – to build a healthy and competent workforce. We recognise that increasing throughput and building graduate competencies remains a challenge for the department and the HEAIDS programme – and these interventions are an important part of the response.

Our plans also aim to build capacity within colleges so that soon, in a sustainable way, they can provide effective and comprehensive student, and staff, support services. The study points to vulnerabilities among TVETs. These institutions have traditionally not been integrated into the health education and service network – which has left TVET students and teachers without much formal support and protection.

Hence the findings of the study will help inform strategies targeted at developing a skilled, capable and healthy workforce – as well as championing transformation imperatives to address social inequality, as well as issues of race, gender, age, geography and disability within the sector.

Despite weaknesses, TVETs hold the potential to fast-track the educational missions because of their excellent geographic spread, diversity of courses and diplomas on offer, links to communities and easy access that our young people have to TVETs.

The study guides us to pursue the following priorities:

- To develop and provide HIV, STIs, TB and general health and wellness interventions to enable our population to implement positive behaviour change in relation to preventable and manageable health conditions.
- To advocate for consistent and effective condom use by students and staff, and address other risky behaviours and socio-structural factors that make them vulnerable to infection.
- To protect the LGBTI community and other people in our sector who are subjected to stigma and discrimination, as well as greater risk of infection and other health problems.
- To implement initiatives that sustain health and wellness of students and staff.

With all this in mind, the findings of this report provide us a better evidence base from which to prepare and implement strategies which will allow young people and staff across TVETs to make choices about health-seeking behaviour and to respond to their own psychological and social needs more confidently.

Mduduzi Manana

Deputy Minister of Higher Education

MESSAGE FROM HEAIDS PROJECT DIRECTOR

Statement by the HEAIDS Director: Dr Ramneek Ahluwalia's message

We embarked on the study of HIV in TVETs in 2014 at about the same time as HEAIDS was expanded to bring these institutions into the fold of our higher education and training HIV, AIDS, STIs, TB and holistic wellness programme. From having a constituency of a considerable high number of students and staff at 23 higher education institutions (HEIs), we rapidly grew into one with two million beneficiaries spread across 23 HEIs and 50 TVETs and all of the nine provinces.



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Understandably we were much less familiar with our new family members – students and staff at TVETs – than we were with the established university and universities of technology population.

Historically, the health and wellness of the TVET students and staff received little attention and we anticipated that the programme and services to meet their needs would be deficient. But we also had to have concrete data to ensure an appropriate response. We realise that there is a need to incubate a large health-promoting intervention that can support young people attending TVETs and engage them in a critical manner so they feel enabled to look after themselves.

We can commence work now we have the findings of this study. They are not all-encompassing and in some areas they suggest further questions to explore so we can come up with cost-effective and suitable solutions. These need to take account of factors and limitations within the TVET sector, such as administrative resources and campuses that do not have the large student populations we see, for example, in large universities based in big metros.

The findings of the 2008-9 HIV knowledge, attitudes, behaviours, perceptions study at universities helped HEAIDS formulate the seven dedicated programmes for the university sector: First Things First HIV, TB, STI testing; women health empowerment; men health; LGBTI/MSM; Future Beats youth development; alcohol and drugs prevention; and health curriculum development. Similarly, the findings of this TVET study will now help us in developing a comprehensive strategy for the TVET sub-sector and extending the programmes to this important population.

This survey explored HIV/AIDS and related factors that affect the sector and which need to be addressed in order to mitigate the impact of the HIV and TB co-epidemic within it:

- The overall knowledge, attitudes and behaviours towards HIV/AIDS in the TVET sector.
- Knowledge about specific aspects of HIV/AIDS and sources of information about it.
- Students' attitudes towards stigma.
- Sexual behaviours and the scope of risky behaviours that may increase rate of HIV infections.
- The effectiveness of existing HIV/AIDS programmes at TVETs.

The data helps us envisage the extent of the challenge. Most of those who participated in this study seem to have absorbed key facts about HIV prevention and care and treatment for people living with AIDS. But many cannot pinpoint where they acquired this information. When we dig a little deeper, many respondents were not certain about the details of, for example, post-exposure prophylaxis (PEP), an essential service for rape survivors. The large number of first-year students who said that they were either pregnant or had made another person pregnant and that vast majority of these were not planned pregnancies suggests that promoting and making contraceptive solutions easily accessible is an urgent priority. Generally, the study participants had positive and progressive views about HIV/AIDS, but there is a group that holds certain prejudices that could slow down the pace of transformation and better health outcomes.

So there is work to be done and we trust that this publication will help all of us working with TVETs in this quest.

A handwritten signature in black ink, appearing to be 'R. Ahluwalia', written over a horizontal line.

Dr Ramneek Ahluwalia
Programme Director

The first national survey on HIV and AIDS knowledge, attitudes and behaviour among students and staff at Technical and Vocational Education and Training (TVET) colleges would not have been possible in such a short space of time without the involvement of several individuals and organisations. We would like to thank the following colleagues and participants who ensured that the research went well within the defined time frame:

The HEAIDS project team: We thank Dr Ramneek Ahluwalia, Director and Head of the Higher Education and Training HIV and AIDS Programme (HEAIDS) for his vision and foresight in commissioning this work. Other members of the team we particularly wish to thank are: Ms Managa Pillay, Senior Programme Manager for Curriculum Development and Research, Ms Sinikiwe Sithole, Senior Programme Manager for Programmes, Mr Alex Semba, Programme Manager for Higher Education Institutions, Ms Brenda White, Project Manager for TVET Support, and Ms Felistas Chikazunga, Senior Finance and Procurement Officer.

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SACPO: Acknowledgement for opening the TVET colleges where this survey was undertaken. We look forward for a more sustained relationship as we implement findings and recommendations for future implementation. Hand full thanks for Ms Hellen Ntlatheng, the current SACPO president.

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Special thanks are due to the **principals, managers, administrators, lecturing staff, union representatives and student leaders** at participating TVET colleges for their assistance and support during the recruitment and data collection processes.

No research study of this nature can be successfully implemented without the support of respondents. We thank the **staff and students at TVET colleges** who participated in the study enthusiastically.

We are grateful to the **financial partners**, USAID for support through the Sexual HIV Prevention Programme (SHIPP) and Dr John Imrie – and the Global Fund to Fight AIDS, TB and Malaria which funded the project through the Networking HIV, AIDS Community of South Africa (NACOSA)

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LIST OF ACRONYMS AND ABBREVIATIONS

AIDS	Acquired immune deficiency syndrome
ART	Antiretroviral treatment
ARV	Antiretroviral (medicine)
DHET	Department of Higher Education and Training
HAST	HIV and AIDS, sexually transmitted infections and tuberculosis
HCT	HIV counselling and testing
HEAIDS	Higher Education and Training AIDS Programme
HEI	Higher education institution
HESA	Higher Education South Africa
HIV	Human immunodeficiency virus
HR	Human resources
HSRC	Human Sciences Research Council
KAB	Knowledge, attitudes and behaviours
KABP	Knowledge, attitudes, beliefs and practices
LGBTI	Lesbian, gay, bisexual, transgender and intersex
MMC	Medical male circumcision
MRC	Medical Research Council
MSM	Men who have sex with men
NACOSA	Networking HIV and AIDS Community of South Africa
NSP	National Strategic Plan for HIV, TB and STIs, 2012-2016
PEP	Post-exposure prophylaxis
PHC	Primary healthcare
PEPFAR	President's Emergency Plan for AIDS Relief
PMTCT	Prevention of mother-to-child transmission
REC	Research ethics committee
SANAC	South African National AIDS Council
SBCC	Social and behaviour change communication
SHIPP	Sexual HIV Prevention Programme
Stats SA	Statistics South Africa
STI	Sexually transmitted infection
TB	Tuberculosis
TVET	Technical and vocational education and training (college)
UNAIDS	Joint United Nations Programme on HIV and AIDS
USAID	United States Aid programme
VCT	Voluntary counselling and testing
WHO	World Health Organization

Introduction

The purpose of the Higher Education and Training HIV and AIDS Programme (HEAIDS) is to reduce the threat of HIV in the higher education sector and to mitigate its impact. In 2009, HEAIDS commissioned a pioneering study on HIV prevalence and knowledge, attitudes, beliefs and practices (KAPB) at the country's 23 public universities. The present national survey of knowledge, attitudes and behaviours (KAB) of students and staff at technical and vocational education and training (TVET) colleges was the first to be conducted in this sector and served to obtain baseline measures on knowledge, attitudes and behaviour pertaining to HIV and related risk factors.

Study methodology

HEAIDS conducted the KAB study in the TVET sector through the Human Sciences Research Council (HSRC).

The study population consisted of 657 690 students and more than 16 000 academic and non-academic staff at 50 TVET colleges across the country. The colleges are housed in 236 campuses in both rural and urban areas. Field work took place during the months of June to September 2014 and involved all 50 TVET colleges.

The study utilised a multi-stage cluster sampling methodology to select a sample of campuses and classes within campuses. Only first-year students were considered for selection and there was no sampling of individual students; the class was the final unit of analysis. The staff sample included academic, management and service personnel. A national sample of 4 500 first-year TVET students and 900 staff members – stratified by academic, management and service occupations – was drawn.

The study was approved by the HSRC's Research Ethics Committee (REC) and permission to conduct the study was obtained from relevant authorities at TVET colleges.

Findings

Demographic characteristics of respondents

Data was collected from 6 654 participants, 5 651 (85%) of whom were students and 1 003 (15%) were staff. The response rate exceeded the intended sample by 23%.

Among students, 78% of respondents were aged 18-24 years, 13% were older than 24 years and 9% were younger than 18 years. There were slightly more female (51%) than male (49%) respondents with a race composition of 87% African, 6% white, 6% coloured and 1% of Indian origin. More than half (53%) of respondents were Christian, one-third were African Traditionalists, 2% were Muslim, 1% were Jewish and 11% adhered to other unspecified religions.

Knowledge of HIV and AIDS, STIs and TB

Awareness of AIDS among students and staff at TVET colleges was high: 87% of the respondents said they had heard of AIDS. Awareness was significantly higher among students (90%) than staff (71%).

Knowledge about HIV transmission was very high, with 96% of students and staff knowing the virus could be passed on through unprotected sex. A high proportion of students (84%) and staff (90%) knew that sexually transmitted infections (STIs) put people at greater risk of HIV infection. However, 37.9% of students and 25% of staff admitted that they did not know whether having anal sex increased the risk of HIV infection.



More staff (82.5%) than students (71%) knew that the HI virus could be passed from a pregnant mother to her unborn child, if the mother was infected. Somewhat surprisingly, only 61.6% of students and 72% of staff knew that people could protect themselves from acquiring HIV by not having sexual intercourse.

More than two-thirds (70%) of students and 79% of staff were aware that AIDS could not be cured. However, 16% of the combined sample still believed that AIDS could be cured.

There were relatively high levels of awareness and knowledge about biomedical methods of prevention HIV. About four in five respondents (81%) said they had heard about medication that HIV-positive pregnant women could take to reduce the risk of infecting the baby with HIV. Only 60% of students and 75% of staff had heard about medication that could help reduce the risk of HIV infection if a woman had been raped.

Nearly one-third of students and staff had undergone TB testing and about 87% of the respondents were correctly informed that TB could be cured.

Attitudes to people living with HIV

Most students and staff at TVET colleges had positive attitudes to people living with HIV (PLHIV) but a minority in both groups were unwilling to be associated with or share living space with PLHIV.

More students (21%) than staff (14%) said they did not want to be associated with HIV-positive people and more male students (16.1%) than females (11%) felt this way, as did more white students (25%) than Africans (13%).

Similarly, more students (24%) than staff (19%) said that sharing a house with HIV-positive people would be very difficult for them. More male students (25%) and male staff (22%) felt this way than female students (22%) and female staff (17%).

More students (20%) than staff (14%) said that people who became infected with HIV were promiscuous. More male students (22%) than female students (17%) felt that HIV-negative people should not be allowed to socialise with HIV-positive people.

Attitudes to HIV and AIDS treatment

Students and staff at TVET colleges had positive attitudes to treatment for HIV and AIDS. A similar proportion of students (79%) and staff (77%) said they trusted that HIV treatment worked.

Approximately 80% of students and 82% of staff agreed that HIV treatment would keep an HIV-positive person alive. More than eight out of 10 respondents rejected the notion that antiretroviral (ARV) medication was poisonous.

Attitudes to condom use and HIV and STI prevention

A substantial proportion of students and staff expressed negative attitudes to condom use. For example, 40.8% of students and 43% of staff said that condoms felt unnatural, and nearly one-third of students and staff said that condoms changed the climax.

More than a quarter of students (26%) and staff (27%) said they used condoms only if their sexual partners wanted them to.

There appeared to be uncertainty about aspects of STI prevention among both students and staff. About four out of 10 respondents said they didn't know what their partner's response would be if they were to ask him/her to undergo an STI test. In addition, 37% agreed or strongly agreed that if they asked their main sexual partner to get tested for STIs the partner might suspect the respondent had cheated. About a third agreed or strongly agreed that if they asked their main sexual partner to get tested for STIs the partner might feel s/he was being accused of giving the respondent an STI.

Sexual orientation and related attitudes

A high proportion of students (83%) and staff (87%) identified themselves as heterosexual, while 7% of students and 5% of staff said they were homosexual and 5% of students and 3% of staff described themselves as bisexual.

Vaginal sex was most frequently selected as the preferred sexual option for both students (72%) and staff members (68%). About one in five students and staff preferred oral sex, and one in 10 chose anal sex.

Only 57% of students and 70% of staff said they felt safe from physical harm on campus. This means that a substantial minority felt insecure about their physical safety while at college.

When asked about the safety of the college environment for lesbian, gay, bisexual, transgender and intersex (LGBTI) students, 53% of students and 61% of staff said they considered their institution to be a safe environment. A slightly higher proportion felt their college was an accepting environment for LGBTI. However, only 22% of students and 42% of staff said management of the institution took discrimination against the LGBTI community seriously.

Behaviours related to HIV risk and risk-reduction

Approximately 61% of students and 72% of staff at TVET colleges had sexual partners and 15% of respondents had sexual partners who were five or more years older than themselves, while 10% had partners who were five or more years younger than themselves.

Condom use at last sex was low when this was with a regular partner or a sex worker. More students (55%) than staff (35%) said they had used a condom at last sex. Consistent condom usage ("every time" they had sex) was reported by 57% of students and 52% of staff members. It is somewhat surprising that the reported rate of consistent condom use was higher than the reported rate of use at last sex. Condom use was highest among both students and staff (76% for both groups) with a non-regular partner.

The self-reported rate of male circumcision was high: almost two-thirds of students (66%) and over half of staff (54%) said they had been circumcised. Medical circumcision performed in a hospital was preferred by 76% of students and 85% of staff while traditional circumcision was supported by 18% of students and 12% of staff. There was overwhelming support for medical male circumcision: 86% of staff and 79% of students said they preferred medical circumcision to traditional circumcision for themselves or their male partners.

About 59.5% of student respondents and 57.5% of staff members indicated that they consumed alcohol. Use of other habit-forming substances at TVET colleges was mainly confined to marijuana or dagga (11% of students used it and 5% of staff). However a small proportion of students and staff indicated use of cocaine or crack cocaine (2% of students and 1% of staff) and mandrax, tik, nyaope and wonga (each of which was used by 1% of students and 0.6% of staff members).

One in 20 students and a slightly higher proportion of staff members said serious crimes like murder, shooting and rape had occurred at their institutions. Assaults on campus were even more frequently reported, by 17% of students and 21% of staff members. Fights involving weapons were said to have taken place at TVET colleges by 12% of students and 9% of staff members. About 14% of students and 8% of staff said they were aware of gangs operating at their institution.

Students and staff presented contrasting pictures in relation to pregnancy-related experiences. Among staff, 70% of respondents said they had been pregnant or made someone pregnant and two-thirds of these pregnancies were planned while 10% had led to an abortion.

Among students – all in their first year of study – 31.7% had been pregnant or made someone pregnant and only a small proportion of these pregnancies (25%) had been planned. Among students, 16% said they or a sexual partner had had an abortion.

Existing programmes at TVET colleges

Students and staff both relied overwhelmingly on government health facilities for AIDS-related health services. Seven in 10 students and staff obtained their services from government health facilities, 11% of students and 35% of staff from medical doctors, and 8% students and 32% of staff from private health facilities.

While staff members used the mass media quite extensively for HIV and AIDS information, only a minority of students did the same. For example 68% of staff members used TV, 59% radio and 48% newspapers to access information on HIV, while the figures for students were 20% for TV, 20% for radio and 17% for newspapers

Online and social media were used much less frequently than traditional mass media to obtain information on HIV.

Overall, TVET colleges did not rank highly as a source of information on HIV for their students and staff. TVET colleges in some provinces were more likely to be mentioned as a source of information. They were Limpopo (32%), Mpumalanga (27%), Western Cape (23%) and Eastern Cape (22%). Only 16% of respondents in the Free State indicated that their colleges provided information on HIV.

Distribution of condoms and HIV testing were reported to take place on a minority of campuses, while on-campus healthcare services related HIV, TB and STIs were generally unavailable.

Conclusions

The positive findings from this study were:

- Awareness of AIDS among students and staff at TVET colleges was high (87% of respondents said they had heard of AIDS) as was awareness and knowledge about medical treatment for AIDS.
- Knowledge about HIV transmission among students and staff was also high and the vast majority of respondents knew where to obtain condoms.
- Students and staff at TVET colleges generally had positive attitudes to PLHIV, treatment for HIV and AIDS, and LGBTI individuals. However, negative attitudes in all three areas persisted among a small minority.
- Condom use at last sex was high with a non-regular partner and the self-reported rate of male circumcision was high. There was general support for circumcision being performed in hospital by medical doctors.
- Substance use other than alcohol was low among students and staff at TVET colleges.

The negative findings from this study were:

- A minority of students and staff at TVET colleges had negative attitudes to sharing with HIV-positive people.
- Attitudes to condom use were negative and many respondents indicated that they only used a condom if their partner requested this.
- Among students and staff there was uncertainty and fear of an adverse partner reaction to control of STIs through partner testing.
- Condom use at last sex was low except with a non-regular partner.
- Overall, alcohol use at TVET colleges was high among students and staff at just under 60%. The rate was considerably higher among some sub-groups – for example, male African students in Gauteng and KwaZulu-Natal.
- Students showed an unusually low interest in the mass media as a source of health information and social media have not taken off strongly as sources of HIV information at TVET colleges.

Recommendations

It is recommended that HEAIDS, in partnership with TVET colleges, should develop appropriate programmes to build knowledge and shift attitudes and behaviour in accordance with the above findings. These interventions should include, but not necessarily be limited to, the items mentioned below.

- Introduce or strengthen the *First Things First* HIV counselling and testing (HCT) campaign for students and staff at all TVET colleges and ensure it incorporates screening for TB.
- Develop *innovative information and education programmes*, using relevant technology platforms and catering to the distinct needs of students and staff.
- Develop *behaviour change interventions* to help students and staff at TVET colleges to respect the rights of PLHIV, LGBTI and women.
- Introduce *coloured and scented condoms* at TVET colleges and address attitudes among students and staff that impact adversely on consistent condom use with *all sexual partners*.
- Introduce interventions to increase knowledge and change attitudes about *prevention and effective treatment of STIs* among students and staff.
- Introduce programmes to promote *responsible alcohol consumption* targeting sub-groups of students with particularly high consumption.
- *Build a sector-wide multidisciplinary response* to HIV and AIDS, STIs and TB that incorporates student leadership, staff unions and management structures and encourages widespread participation of the TVET community.



1.1 Background on Technical and Vocational Education and Training colleges

South Africa has 50 technical and vocational education and training (TVET) colleges which had a total of 657 690 registered students in 2013, according to the 2014 report on post-school training. These colleges are distributed across nine provinces and they are accommodated in about 300 campuses situated in rural and urban areas.

TVET students ranged in age from 15 years to over 40 years, but the vast majority enrolled in 2013 (92%) were aged 18 years and older. Males accounted for 52% of enrolled students and the racial distribution was estimated as follows: 80% African, 9% coloured, 7.8% white, 2.1% of Indian origin, and 2.3% belonging to other categories (Department of Higher Education and Training, 2014).

Courses offered at TVET colleges include those accredited by the Services SETA and courses in accounting, business, computer science, management, human resources practice, project management, event management, beauty therapy, secretarial practice, décor and design, sports and fitness, forensic investigation, hospitality, writing, wedding planning, and child care.

The introduction of HIV and AIDS programmes at TVET colleges has been slower than at other higher education institutions (HEIs). However, several programmes are now operational at TVET colleges and they include:

- A key annual campaign of the Higher Education and Training HIV/AIDS Programme (HEAIDS), *First Things First*, which encourages students at tertiary institutions – and especially incoming first-year students – to take responsibility for knowing their HIV status. At some TVET colleges this campaign runs under the name, *Shout it Now*.
- *HEAIDS Men Health Empowerment Programme*, an initiative of the Centre for Communication Impact (CCI, known previously as Johns Hopkins Health and Education in South Africa or JHHESA). This is implemented in partnership with HEAIDS and is directed at men. It promotes healthy lifestyles for men, deals with masculine sexual norms, and encourages medical male circumcision, condom use, being faithful to a single partner, and responsible use of alcohol.
- *HEAIDS Women Empowerment Programme*, another CCI-HEAIDS partnership, which aims to empower young women and addresses behaviours, such as gender-based violence and transactional sex, which drive HIV risk.
- A programme that seeks to prevent alcohol and substance abuse, which is a risk factor for HIV infection, and provides professional assistance to those needing help to overcome substance abuse.
- A programme that strives to develop an enabling environment on campuses for men who have sex with men (MSM) and LGBTI individuals and ensure they have access to HIV-related health services.
- A campus radio pilot project, *Future Beats*, which aims to develop compelling programming on HIV and related topics, including gender violence, substance abuse, human rights and social justice.
- A peer education programme offered at some TVET colleges by loveLife.

1.2 Background on HEAIDS

HEAIDS is an initiative of the Department of Higher Education and Training (DHET) that is implemented through Higher Education South Africa (HESA) in partnership with the South African College Principals' Organisation (SACPO), which represents TVET colleges. The purpose of the HEAIDS programme is to develop and support HIV mitigation programmes at South Africa's public HEIs and TVET colleges.

HEAIDS is overseen by a Strategy Group that includes HIV experts and representatives of HEIs, some TVET colleges and relevant stakeholders. HEAIDS enables HEIs and TVET colleges to respond to the HIV and AIDS epidemics through their core functions of learning and teaching, research and innovation, and community engagement.

More specifically, the programme assists HEIs and TVET colleges to:

- Develop HIV prevention programmes for students and staff and facilities for the treatment, care and support of students and staff living with HIV.
- Provide a comprehensive workplace programme on HIV and AIDS that caters to the needs of staff.
- Educate and equip students to make a contribution to the national HIV and AIDS response in their future careers.
- Conduct research that will strengthen society's ability to resist and ultimately overcome the pandemic.
- Provide HIV and AIDS services to communities through outreach projects and practical training programmes

It has been a long-standing concern of the DHET that the response of many TVET colleges to the HIV epidemic has been inadequate and that the HEAIDS programme should be extended to benefit the TVET sector. In addition, very little evidence has been collected on the impact of HIV on the TVET sector and how colleges have responded to the challenges.

Certain assumptions can be made about the susceptibility of the TVET population to HIV, sexually transmitted infections (STIs) and tuberculosis (TB), given the social and demographic profile of students attending these colleges and the location of many of the colleges.

In January 2014 HEAIDS commissioned a study by the HIV and AIDS, STI and TB (HAST) research programme of the Human Sciences Research Council (HSRC) in order to obtain an overview of the knowledge, attitudes and behaviours (KAB) of students and staff in the TVET sector in relation to HIV and AIDS, STIs and TB. This information was also intended to serve as baseline data for the development of relevant programmes of intervention. The survey was supported financially by the Global Fund to Fight AIDS, TB and Malaria (GFATM) through the Networking HIV and AIDS Community of South Africa (NACOSA) and by USAID through the Sexual HIV and AIDS Prevention Programme (SHIPP).

1.3 HIV and AIDS among young people

HIV prevalence among young South Africans, aged 15-24 years, is one of the world's highest (UNAIDS, 2010) although there has been a decline from 10.3% in 2005 to 8.7% in 2008 and 7.3% in 2012. In 2012, young women in this age group recorded an HIV prevalence rate of 11.4%, which was 3.9 times higher than the rate for their male counterparts (2.9%) (Shisana et al, 2014).

The HSRC 2014 population-based survey reported that there were approximately 469 000 new HIV infections in 2012, which translates to an average of 1 500 new infections a day. One-third of these (192 000) were among young people in the 15-24 years age group. In this age group, females accounted for 90% of new infections. The age group in which the highest rate of new

infections occurs (15-24 years) is also the age group to which most students at TVET colleges belong.

The HEAIDS 2010 study on HIV prevalence and knowledge, attitudes, beliefs and practices (KABP) found that the prevalence rate was about three times higher among students over the age of 25 years than among younger students. Female students were hardest hit, with prevalence of 4.7% which was twice the rate for their male peers (HEAIDS, 2010). The study revealed wide variations in HIV prevalence among HEIs in different regions. Prevalence was highest among students in the Eastern Cape (6.4%) and lowest among those in the Western Cape (1.1%). There were also large variations between universities within a single region.

The National Survey of HIV and Sexual Behaviour among Young South Africans found that the highest HIV prevalence by province was in KwaZulu-Natal (14.1%) and the lowest in Limpopo (4.8%). Young people living in urban informal areas had the highest HIV prevalence (17.4%), followed by rural formal areas (13.5%), urban formal areas (9.8%) and rural informal areas (8.7%) (Pettifor et al, 2004).

1.4 HIV-related knowledge, attitudes and behaviours among young people

1.4.1 Knowledge of HIV and AIDS among young people

Initiatives to improve knowledge about HIV and AIDS centre on providing information about the modes of transmission, means of prevention and behaviours that reduce susceptibility.

A UNAIDS report (2012) stated that most youth did not know the modes of HIV transmission and were poorly informed on methods by which they could protect themselves against infection. The same study revealed that half of all youth in the countries surveyed had mistaken beliefs about the transmission of the virus. The report concluded with two major goals:

- Reducing the number of youth infected with HIV.
- Providing information, education and services to youth across the globe.

In South Africa, recent findings indicated that the proportion of young people aged 15-24 years with comprehensive knowledge of HIV prevention was generally high although there were some gaps. Similarly, attitudes towards people living with HIV were generally positive although some attitudinal barriers were identified (HEAIDS 2010).

According to the National Youth Risk Behaviour Survey (2008), 95% of young people knew that there were ways to avoid HIV infection and 77% knew that use of condoms prevented transmission. The survey suggested that 62% of HIV-infected young people thought they were at little or no risk of contracting HIV. This finding confirmed other studies in which young people appear to underestimate their personal risk of infection (Reddy et al, 2010).

Despite high levels of knowledge of HIV and AIDS among young people, studies have revealed a mismatch between knowledge and reported sexual behaviours (Simbayi et al, 2005; James et al, 2004; Panday et al, 2009). For example, Simbayi et al (2005) examined risk behaviours and HIV risk factors among young people living in a South African township and established that 68% of men and 56% of women reported engaging in sexual behaviours with a high risk of HIV transmission. Although knowledge about HIV transmission was generally high, there was evidence that misconceptions persisted, particularly myths related to HIV transmission. For young men, HIV risk factors were associated with fewer years of education, lower levels of AIDS-related knowledge, negative attitudes to condoms, and dagga (marijuana) use. Among young women, HIV risk factors were high rates of unprotected vaginal intercourse and the perception that condoms get in the way of sex.

Despite the gap between knowledge and behaviour, adequate information is an important factor in changing attitudes and behaviours related to HIV (Pettifor et al, 2004).

1.4.2 Young people's attitudes to HIV and AIDS

For knowledge about HIV transmission to result in risk-reduction behaviour, shifts in attitudes and social norms are often required in order to create environments that support and encourage change among young people (Panday et al 2009; Pettifor et al, 2004).

In a national survey on HIV and sexual behaviour among 15-24-year-olds in 2003, most participants agreed with the statement that safer sex was the shared responsibility of both partners (96%). The majority disagreed that it was acceptable to have many sexual partners (97%), to pressure someone to have sex (98%) or to have a sugar mommy or sugar daddy who bought you things in exchange for sex (95%) (Pettifor et al, 2004).

In the same survey, 20% of young people agreed that it was cool to have a sexual partner who was older. Females were significantly more likely to agree with this statement than males. While young people were confident that they could discuss condom use with their partner (90%), fewer were confident that they could use a condom after drinking alcohol or taking drugs (43%) (Pettifor et al, 2005). It was frequently reported that condom use was discouraged by feelings of trust, respect and fidelity that developed within steady relationships, while condoms were generally accepted in casual encounters (Panday et al, 2009).

The HEAIDS 2010 study found that there was still considerable stigma attached to HIV. Only 38% of students believed they would be supported by their friends if they revealed that they were HIV-positive. Whilst stigmatising attitudes to HIV may actually be low, young people might have negative expectations which are driven by peer influence.

Stigma is often a primary barrier to effective HIV prevention and to provision of treatment, care and support to those living with the virus. For example, a young person might feel uncomfortable to access antiretroviral treatment at a neighbourhood clinic because of fear of being labelled by other young people in his or her social group.¹

A study among young adults in Cape Town found that a vast majority of respondents (97%) had some stigmatising attitudes or negative behavioural intentions to PLHIV. These stigmatising attitudes were moderate in 25% of respondents and high in only 1% of respondents (Maughan-Brown, 2006).

There is ample evidence that communication campaigns and programmes build knowledge relating to HIV. But there is less consistent evidence of their effectiveness in improving skills or changing values, attitudes and peer norms (Panday et al, 2009). There is a need for more rigorous evaluation of institution-based HIV interventions that focuses on biological outcomes, such as STI, HIV and pregnancy rates, which would indicate behaviour change.

1.4.3 Behaviours related to HIV and AIDS among young people

Sexual risk behaviours are defined as sexual activities that may expose an individual to the risk of infection with HIV and other STIs. Examples would include unprotected sex, early sexual debut, multiple sexual partners, using alcohol or drugs before sexual intercourse, forced or coerced sexual intercourse, and sexual intercourse for reward (UNAIDS, 2012).

The HEAIDS 2010 study on KABP at HEIs found that 60% of sexually active students reported using a condom the last time they had sex. About 8% of all sexually active students reported engaging in anal sex and some students mistakenly perceived anal sex as less risky than vaginal intercourse.

The 2nd South African National Youth Risk Behaviour Survey (YRBS) found that 31% of sexually active secondary school students reported that they always used a condom when having sex. Significantly more white (43.2%) and Indian (50.5%) school students said they always used

¹ <http://www.afroaidsinfo.org/ShowProperty?nodePath=/BEA+Repository/files/public/sayouth.pdf>

condoms than African (30%) and coloured (30.7%) participants. Condom use was higher among students in Grade 11 than in Grade 8 (Reddy et al, 2010). In addition:

- More than half of those students who reported having had sex (52.3%) said that they had had one or more sexual partners in the past three months.
- About 16.2% of students reported that they had used alcohol before sex, with more male students (19.5%) reporting this than female students (11.6%). The highest rates of alcohol use were among participants in the Western Cape, where 14% of students also said they used drugs before sex.

A recent nationwide survey on HIV found that 10.7% of respondents aged 15-24 years (16.7% of males and 5.0% of females) said that they had sex for the first time before the age of 15 years (Shisana et al, 2014). In addition, 19.9% of respondents aged 15-19 years were involved with a sexual partner who was more than five years older than they were. One-third (33.7%) of all female adolescents aged 15-19 years reported having had a partner more than five years their senior, compared to 4.1% of their male counterparts. More males (20.1%) than females had multiple partners. Rates for multiple partners were highest (22.4%) in the 15-24 year age group (Shisana et al, 2014).

Studies have shown that young women with older partners are at increased risk of HIV infection. Multiple sexual partners and inconsistent condom use have also been associated with HIV infection among both men and women (Panday et al, 2009).

There has been a decline in HIV prevalence among pregnant women aged 15-24 years in South Africa, from 21.8% in 2010 to 20.1% in 2011 and 19.3% in 2012 (NDoH 2012). But there is a strong association between unplanned pregnancy and HIV infection among young people (Panday et al, 2009) and unplanned pregnancies are also associated with an increase in school drop-out rates (Pettifor et al, 2008).

1.4.4 HIV and AIDS programmes for young people

In view of the high rate of HIV infection among young people in South Africa, programmes to address the sexual behaviour of adolescents have drawn considerable scholarly interest. These include school-based sex education interventions, peer education programmes, adolescent-friendly clinic initiatives, mass media interventions and community-level programmes (Pettifor et al, 2005; Panday et al, 2009).

HEAIDS exists to “develop and strengthen the capacity, systems and structures of all higher education institutions (HEIs) to prevent, manage and mitigate the causes, challenges and consequences of HIV/AIDS” (HEADS, 2010).

The HEAIDS 2010 survey found that only 53% of students, 60% of academic staff, 62% of administrative staff and 54% of service staff felt that their institutional management was taking HIV and AIDS seriously. With a few exceptions, HEIs did not utilise their role as research institutions to conduct relevant research on HIV and AIDS on their own campuses. Many students and staff felt that HIV-related services on campus were lacking or inadequate and this hindered HIV counselling and testing (HCT).

HIV education and awareness campaigns constitute an important part of the South African government’s HIV prevention strategy. AIDS awareness campaigns funded by government include Soul City and the Medical Research Council’s AfroAIDSinfo project. These are programmes which encourage healthy sexual behaviour utilising mass media such as television, radio and the Internet.

Research shows that well-designed programmes, which provide information, motivation and behavioural skills, are effective in delaying sexual debut among adolescents and increasing the

proper use of contraceptives and condoms by those who choose to be sexually active (Brown and Eisenberg, 1995; Kirby et al, 1994). Opponents of sex education claim that providing young people with broadly based sex education will result in young people engaging in sex at an earlier age (McKay, 1993). A number of studies have investigated this issue and no association has been found between exposure to formal sex education and earlier sexual experience (McKay, 1993). Kirby et al (1994) concluded, from their extensive review of sex education programmes, that including discussions on contraception in combination with other topics does not hasten sexual debut.

1.5 Goal and objectives of the study

The goal of this study was to assess knowledge, attitudes and behaviours related to HIV and AIDS in the TVET college sector in South Africa.

The specific objectives were to:

- Assess the level of knowledge of various aspects of HIV and AIDS and identify information sources about HIV and AIDS.
- Examine the attitudes of students and staff at TVET colleges to HIV and AIDS and the extent of stigma related to HIV.
- Assess the sexual behaviour of students and staff at TVET colleges and the extent of risk behaviour in relation to HIV infection.
- Assess the effectiveness of HIV and AIDS programmes that are currently provided in TVET colleges.



2.1. Survey population

The population for this study was 657 690 students and about 16 900 academic and non-academic staff at 50 TVET colleges. These colleges are situated in all provinces, in rural and urban areas, and have a total of 236 campuses. Student participants ranged in age from 15 years to over 40 years. Only 8% of students were younger than 18 years and 52% were male. The racial composition of the student population was estimated at 80% African, 9% coloured, 7.8% white, 2.1% of Indian origin and 2.3% from other categories.

2.2. Survey design

A pre- and post-test study design was developed, with the pre-test survey conducted in 2014 and the post-test component planned for 2016. This kind of study involves comparing the results of two cross-sectional studies on the same population and is suited to measuring the extent of change in knowledge, attitudes and behaviours (KAB) that put students and staff at risk of HIV and AIDS. It would also be appropriate for measuring the efficacy of HEAIDS programme interventions at TVET colleges.

The current report relates to the pre-test survey conducted in 2014. Baseline data was collected on the level of knowledge of students and staff about HIV, their attitudes to HIV-related matters and their behaviours in terms of risk of transmission of HIV. The focus was deliberately on first-year students, the majority of whom were new to college and had not been exposed to any HEAIDS intervention. The data was collected in the period from June to September 2014.

The same data will be collected in 2016 in a study using a sample drawn from the population of third-year TVET college students and staff. This pre- and post-test design will allow researchers to make inferences on the effect of the HEAIDS intervention.

Figure 2.1 depicts the pre- and post-test study design used in this project.

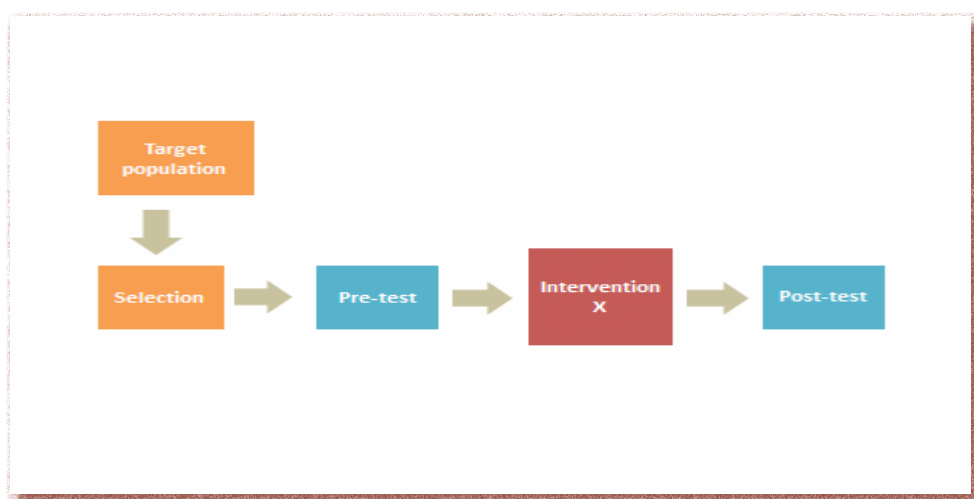


Figure 2.1: Study design used in TVET sector KAB survey

Source: <http://www.nationaltechcenter.org/index.php/products/at-research-matters/quasi-experimental-study/>

2.3. Sampling

The study utilised a multi-stage cluster sampling method to select a sample of campuses and classes within campuses.

All 50 TVET colleges were included in the sampling population in order to represent a range of locality types: rural and urban. Only registered first-year students were considered for selection along with academic, management and service staff. Both students and staff were drawn from all racial groups.

In the selection of student participants, the class was the final unit of analysis – meaning that all students in a selected class were invited to participate in the study. For staff, the campus was the final unit of analysis and thus all staff members at the sampled campuses were eligible to participate.

Figure 2.2 summarises the process of sample selection.

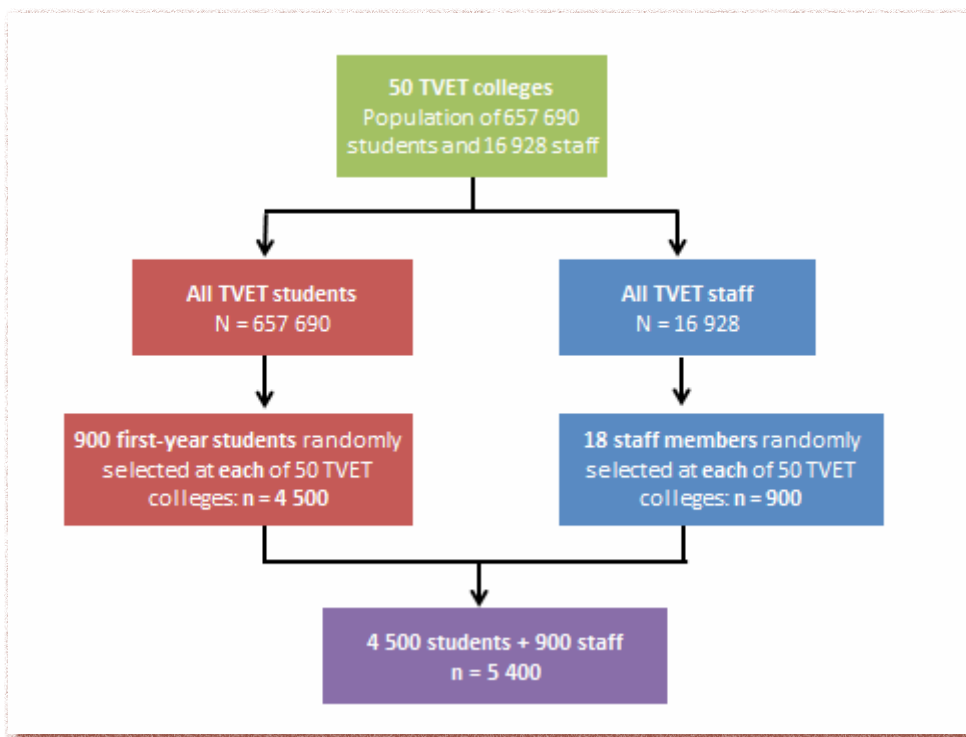


Figure 2.2: Process for selection of participants for KAB survey

A national sample of 4 500 TVET students was drawn, with an average of 500 per province and 90 per institution. A sub-sample of 500 students younger than 18 years was included. These students were equally distributed across provinces.

The use of the class as the final unit of analysis for students catered for participant refusal to participate, absenteeism, and other factors that might result in a low response. It also strengthened the statistical power of the survey.

A total of 900 staff members were sampled from the three main occupational categories – academic, management and service staff – and they constituted 5.3% of all staff working at TVET colleges nationwide.

2.4. Study instruments

The design of two structured questionnaires, for students and staff respectively, was based on previous surveys and informed by a review of literature. The questionnaires included questions on demographic characteristics, HIV and AIDS knowledge, attitudes and behaviours, and exposure to programmes dealing with matters such as HIV counselling and testing (HCT) and care, support and treatment for people living with HIV.

The starting point in designing the data-collection tool was the questionnaires used for staff and students in the HEAIDS 2010 survey of HEIs. Many questions from this earlier study were retained in the current study in order to facilitate comparison of findings.

To increase the validity of the tool and the reliability of the questions, Section A to Section E were based on various publically accessible survey tools, such as the Study on the Determinants of Demand and Supply of Educators based in Public Educational Institutions and FET Colleges (2004/5); the AIDS Indicator Survey (AIS) model individual questionnaire, 2011; the South African National Health and Nutrition Examination Survey, 2011/12; the South African National Health, Demographic and Behaviour Survey, 2011; and the Uganda Demographic Health Survey, 2012.

The section on TB included questions to measure knowledge of the disease, attitudes (including stigma), care-seeking behaviour and sources of information. They were developed from the South African National Health and Nutrition Examination Survey (SANHANES 2011/2012) adult questionnaire. TB screening questions were sourced from the Centers for Disease Control and Prevention (CDC) surveys repository and from other similar questionnaires.

2.5. Data collection

2.5.1. Selection and training of field staff

Fieldworkers who had previously worked at the HSRC were selected and hired as data collectors. Training of the data collectors took place at HSRC centres in Cape Town, Pretoria, Durban and Port Elizabeth and covered standardised operating procedures for conducting the survey. The training manual was provided to fieldworkers for use as a reference document in the field.

Table 2.1: Distribution of study sample and fieldwork staff

Province	Staff	Students	Total sample	Number of teams	Number of field staff
Eastern Cape	70	350	420	2	6
Free State	92	460	552	1	4
Gauteng	211	1 055	1 266	3	12
KwaZulu-Natal	175	875	1 050	3	12
Limpopo	85	425	510	2	7
Mpumalanga	56	280	336	1	3
North West	56	280	336	2	7
Northern Cape	49	245	294	1	4
Western Cape	106	530	636	2	7
Total	900	4 500	5 400	17	62

2.5.2. Fieldwork and survey administration

Fieldwork was co-ordinated by HSRC researchers who were allocated to each of the nine provinces. The survey was conducted in English although participants were drawn from all language groups represented in South Africa. Students and staff at TVET colleges were sufficiently fluent in English to understand the language used in the questionnaires.

Student questionnaires were administered by data collection facilitators to a group of first-year students in a classroom. The facilitator put the questions to the group by reading from the questionnaire and students ticked their answers on individual questionnaires without consulting other students. The staff questionnaire was administered by the data collector during a one-on-one interview or, where appropriate, it was self-administered.

2.5.3. Data capturing and management

Data were captured in-house at the HSRC by the Data Capturing Unit (DCU) which undertakes this function for all major surveys performed by the HSRC. The data were double captured from paper-based questionnaires using software known as the Census and Survey Processing System (CSPPro).

2.6. Plan for data analysis

Data analysis for the survey was conducted using STATA 12/13 (Stata Corporation, College Station, TX, USA). Descriptive analysis was used to summarise the socio-demographic and socio-economic characteristics of both students and staff. Frequency tables were constructed to show differences in knowledge and attitudes about HIV, TB and STIs among students and staff. The Pearson Chi-square test was used to test for significant differences in responses between students and staff at $p < 0.05$.

2.7. Ethical considerations

2.7.1. Ethical procedures

The study protocol and research instruments were approved by the HSRC's Research Ethics Committee (REC). The reference for their approval is REC 09/19/02/14. The research team also obtained permission to conduct the study from the office of Director-General for Higher Education and Training and from campus principals of all TVET colleges.

2.7.2. Risks to participants

In order to minimise social risks, interviews were conducted prior to the initiation of fieldwork with relevant stakeholders. Facilitators stressed the voluntary nature of the survey and explained the purpose for which the findings would be used. The outcomes of these discussions were used to adjust and guide the execution of the study. Although this was a low-risk study, there were some minor social risks. For example:

- A participant's response might be read or overheard by another participant and this could result in stigmatisation. In order to minimise such risks, students were seated relatively far apart during group survey sessions and individual interviews were conducted out of earshot of others. In addition, completed questionnaires were placed in envelopes that were sealed afterwards and sent directly to HSRC to ensure no unauthorised person had access to them.
- Questions dealing with sensitive issues such as forced sex, HIV status and HIV transmission might result in participants experiencing temporary emotional disturbance. In order to minimise this, members of the field staff were thoroughly trained to respond to such reactions, for example by stopping the interview for a while, listening to the participant, discussing the issue, calming the participant down, and continuing the interview only if the participant agreed to do so.

In addition, participants could refuse to answer any question.

Research staff provided referrals to local services for counselling, care and treatment, as appropriate. In addition, all participants were given the name and telephone number of the local study coordinator should they want to ask questions about the study or believe they were psychologically harmed or poorly treated as a result of inclusion in or exclusion from the study.

It was recognised that first-year students, who had just started college, might feel obliged to participate in the survey. To offset this, information about the study in posters and flyers emphasised that participation was entirely voluntary and the information was confidential. Life orientation lecturers at the colleges also stressed these aspects of the study.

2.7.3. Benefits to participants

There were no direct benefits related to participating in the study.

2.7.4. Voluntary participation

Participants were informed that their participation in this study was strictly voluntary and that they were free to withdraw from the study at any time.

2.7.5. Informed consent (and assent by respondents under 18 years)

After carefully explaining the study, researchers gave eligible participants the consent form to read or read the form to participants, where necessary.

They addressed all questions that prospective participants asked. Participants stated orally that they understood and agreed to all aspects of the consent form. Following this, the participant and a member of the research team both signed the consent form. This process was followed in group sessions as well.

2.7.6. Protection of privacy of individual

A private office space was used for the staff interviews. The questionnaire was administered face to face with no person in the space other than the interviewer and the participant.

Students completed the questionnaire in group sessions organised in their classroom. The seating of students was spread out to ensure that there was no risk of communication or reading over each other's shoulders. All interviews were conducted under close supervision of a trained fieldworker.

2.7.7. Protection of confidential information

No names or personal identifiers of participants were recorded on survey forms. Signed consent forms were stored in a locked filing cabinet in a lockable office, separately from other study data. There was no way of linking names on consent forms to interview forms.

In addition, to assure participants of the confidentiality of their information, the data collection process was done by external HSRC fieldworkers. Because participants and fieldworkers were not familiar with each other, the possibility of participants' information being shared with others was negligible. This approach appeared to reduce the number of refusals to participate.



This chapter describes the findings of the survey in respect of both students and staff members. The results are presented together for both categories of participants and, where appropriate, comparisons are made between their results.

3.1. Demographic characteristics of respondents

All 50 TVET colleges nationwide participated in the study. They were represented by 166 campuses, which constituted 70.3% of the total number of campuses.

In the North West and Northern Cape 100% of campuses were included. The lowest coverage of campuses was in KwaZulu-Natal, where 36 of 57 campuses (63.2%) participated in the study.

Table 3.1: Distribution of TVET colleges and campuses by province

Province	Total number of colleges	Number of colleges visited	Colleges visited (%)	Total number of campuses	Number of campuses visited	Campuses visited (%)
Eastern Cape	8	8	100	32	21	66
Free State	4	4	100	20	14	70
Gauteng	8	8	100	39	28	72
KwaZulu-Natal	9	9	100	57	36	63
Limpopo	7	7	100	18	17	94
Mpumalanga	3	3	100	15	14	93
North West	3	3	100	11	11	100
Northern Cape	2	2	100	7	7	100
Western Cape	6	6	100	37	18	49
Total	50	50	100	236	166	70

A total of 6 654 respondents took part in the survey. Of these, 5 651 (84.9%) were students and 1 003 (15.1%) were staff. The sample size exceeded the number originally planned by 23.2%. This was due to good preparatory work, including extensive advertising of the survey, and a positive response from students in the sampled classes.

Among students, a large majority (77.6%) were in the age group 18-24 years, while 13.2% were older than 24 years and 9.2% were younger than 18 years.

There were slightly more female participants (51.4%) than male participants (48.6%). The racial composition of the sample was 87.1% African, 6% white, 5.9% coloured and 1% of Indian origin.

More than half of the respondents (52.7%) described themselves as Christians, 33.8% were African Traditionalists, 1.5% Muslim and 0.9% Jewish. A total of 11.4% subscribed to other religions. Most student (64.0%) and staff (64.9%) respondents came from urban formal communities, while 21.2% of students and 23.0% of staff were from rural settlements.

Table 3.2: Demographic characteristics of students and staff

Variables	Students		Staff		Total	
	n	%	n	%	n	%
Age						
<18 years	522	9.2	0	0	522	7.8
18-24 years	4 386	77.6	104	10.4	4 490	67.5
>24 years	743	13.2	899	89.6	1 642	24.7
Gender						
Male	2 774	49.1	461	46.1	3 235	48.6
Female	2 877	50.9	540	53.9	3 417	51.4
Race						
African	5 017	88.9	764	76.7	5 781	87.1
White	274	4.9	122	12.2	396	6
Coloured	298	5.3	96	9.6	394	5.9
Religion						
African Traditional	1 988	36.7	178	17.9	2 166	33.8
Christian	2 700	49.8	679	68.3	3 379	52.7
Jewish	51	0.9	5	0.5	56	0.9
Muslim	80	1.5	18	1.8	98	1.5
Other	604	11.1	114	11.5	718	11.1
Locality type						
Rural	1 183	21.2	223	23	1 406	21.5
Urban formal	3 573	64	628	64.9	4 201	64.2
Urban informal	823	14.7	117	12.1	940	14.4

Table 3.3 describes the living arrangements of students and staff. Perhaps predictably, more students (45.7%) than staff (22.9%) were in a relationship but not living with their partner. Approximately 6% of students and 8.3% of staff were living with a boyfriend, girlfriend or partner. About one in five students (18.9%) lived with friends, peers or fellow students, while few staff members did so. More students (22.8%) than staff (16.8%) were living alone. Half of the staff population (50.9%) lived with their husband or wife, while only 6.5% of students shared this situation.

Table 3.3: Living arrangements of students and staff at TVET colleges

Variables	Students		Staff		Total		Test statistic
	n	%	n	%	n	%	
Current living status							
In relationship but not living together	1149	45.7	161	22.9	1310	40.7	$\chi^2 = 866.6$ ($p < 0.001$)*
Living with boyfriend, girlfriend or partner	151	6.0	58	8.3	209	6.5	
Living with friends, peers or fellow students	476	18.9	8	1.1	484	15.1	
Living with husband or wife	164	6.5	358	50.9	522	16.2	
Living alone	573	22.8	118	16.8	691	21.5	

* Pearson chi square

Table 3.4 shows that there was no significant difference between male and female students in terms of the pattern of their relationships ($p > 0.05$). Most male students (82.2%) and female students (83%) were single; 11% of males and 10.9% females were going steady; 4% of males and 3.3% of females were living together but not married; and fewer than 1% of female and male students were married, divorced or widowed.

Table 3.4: Relationship status of students by gender

	Male		Female		Test statistic
	n	%	n	%	
Relationship status					
Single	2 219	82.3	2 338	83	$\chi^2 = 1.75$ $p = 0.781$ *
Going steady	294	11.0	308	10.9	
Living together not married	104	4.0	93	3.3	
Divorced/widowed	15	.002	12	.004	
Married	62	.02	65	.02	

* Pearson chi square

Table 3.5 shows economic situation of students and staff.

A substantial proportion of students said they could not afford the basics of food and clothing (28.6%) or lacked “many other things” (43.8%). About a quarter of students (23.2%) said they had “most important things but few luxury goods”, while only 4.4% said they had money for luxuries.

A slightly different picture emerged for staff. While 12.4% of staff members said they had insufficient money for food and clothes, another 39.7% said that, while these items were covered, they went short of many other things. Nearly four out of 10 staff members said they could afford most important things (but few luxuries), while 10.2% said they had money for “extras” of a luxury nature.

Only 44.6% of student respondents and 41.0% of staff members indicated that they were covered by medial aid. Among this group of students, 70.8% were covered by the medical aid

of a parent, guardian or spouse, 16.5% had their own comprehensive cover and 12.7% had their own hospital plan.

Three-quarters of the sub-set of staff members who confirmed that they had medical aid said they had their own comprehensive medical aid, while the remainder said they depended on a hospital plan (6.1%) or the medical aid of a parent, guardian, spouse or partner (18.2%).

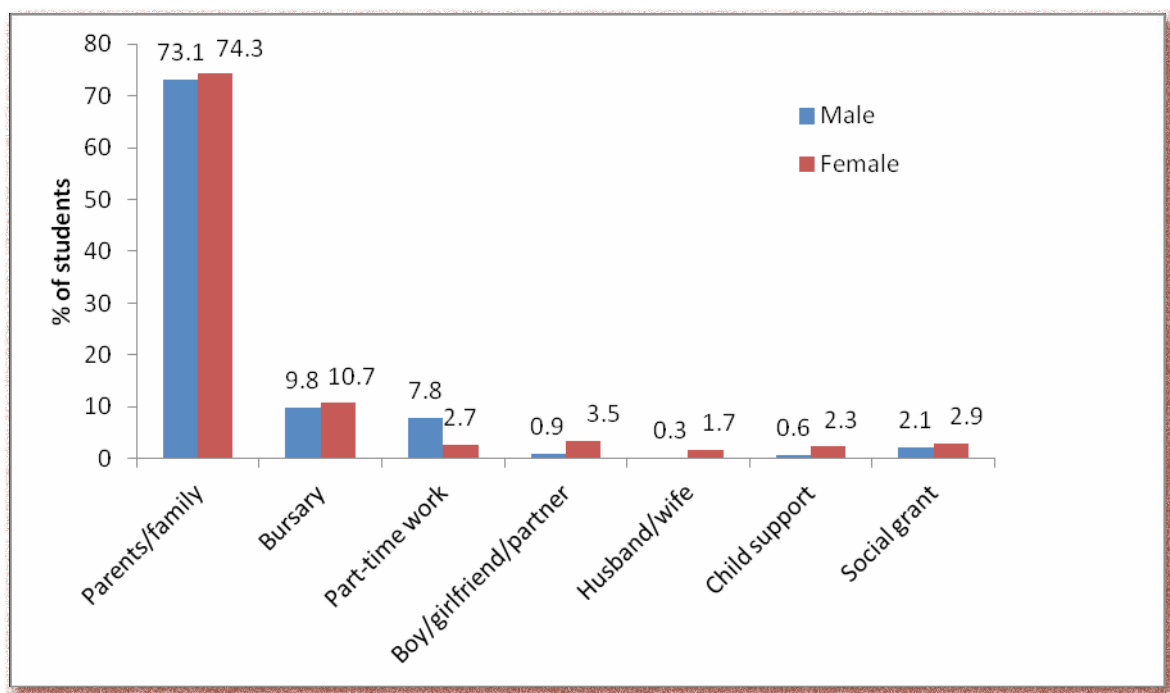
Table 3.5: Socio-economic status of students and staff at TVET colleges

	Student		Staff		Total	
	n	%	n	%	n	%
Financial situation of household						
Not enough money for basic things like food and clothes	1 594	28.6	121	12.4	1 715	26.2
Have money for food and clothes, but short of many other things	2 441	43.8	387	39.7	2 828	43.2
We have most of the important things, but few luxury goods	1 292	23.2	368	37.7	1 660	25.3
We have some money for extra things such as going away for holidays and luxury goods	246	4.4	100	10.2	346	5.3
Medical aid arrangements						
I have my own medical aid (comprehensive cover)	416	16.5	311	75.7	727	24.8
I have my own medical aid (hospital plan)	321	12.7	25	6.1	346	11.8
I am covered by the medical aid of my parent/guardian/partner/husband/wife	1 785	70.8	75	18.2	1 860	63.4

* Pearson chi square

Figure 3.1 shows that the main sources of income for students were parents or family members (71.6%), bursaries (10.6%), part-time work (5.3%), a boyfriend/girlfriend/partner (2.4%), husband or wife (1.2%), social grants (2.6%) or child support (1.5%).

Figure 3.1: Main sources of income for students by sex



3.2. Knowledge of HIV and AIDS, TB and STIs among students and staff

3.2.1. Awareness of AIDS

Overall, 87% of all respondents said they had heard of AIDS. Awareness was significantly higher among students (89.6%) than among staff (70.6%) ($\chi^2 = 325.0$; $p < 0.001$) (Table 3.6).

Table 3.6: Students and staff who had ever heard of AIDS

Ever heard of an illness called AIDS	Students		Staff		Total		Test statistic
	n	%	n	%	n	%	
Yes	4 919	89.6	681	70.6	5 600	87.0	$\chi^2 = 325.0$ $p < 0.001^*$
No	571	10.4	264	27.4	835	13.0	

* Pearson chi square

3.2.2. Knowledge about HIV and STI transmission

Knowledge about HIV transmission among students and staff was high: 96.3% of students and 96.6% of staff knew that HIV could be spread through unprotected sex (Table 3.7).

Most of the respondents – 89.7% of students and 92.7% of staff – knew that people could reduce their risk of acquiring HIV by using a condom each time they had sex.

There was a high level of knowledge (83.8% among students and 89.7% among staff) that the presence of sexually transmitted infections (STIs) puts people at greater risk of HIV infection or reinfection with different strains of the virus. There were similar levels of knowledge that a person who does not appear sick may have HIV infection and pass it to others.

There was less widespread understanding of the need for HIV-positive individuals to protect themselves from reinfection, the specific risks of unprotected anal sex, mother-to-child transmission, and the benefits of sexual abstinence. The survey found that:

- The fact that protection is necessary even when both partners are HIV-positive was known by 77.4% of students and 83.5% of staff.
- Only 48% of students and 61.4% of staff said that having anal sex increased the risk of HIV infection. More than a third of students (37.9%) and a quarter of staff (25.4%) said they did not know whether having anal sex increased the risk of HIV infection.
- More staff (82.5%) than students (71.3%) knew that HIV infection could be passed from an HIV-positive woman to her unborn child.
- Only 61.6% of students and 72.3% of staff knew that individuals could protect themselves from becoming infected with HIV by abstaining from sexual intercourse.

More than two-thirds of students (69.8%) and 78.8% of staff correctly rejected the notion that AIDS can be cured. However, nearly one in three students believed the fallacy that AIDS can be cured or were uncertain whether this was the case. The corresponding figure for staff members was 21.1%.

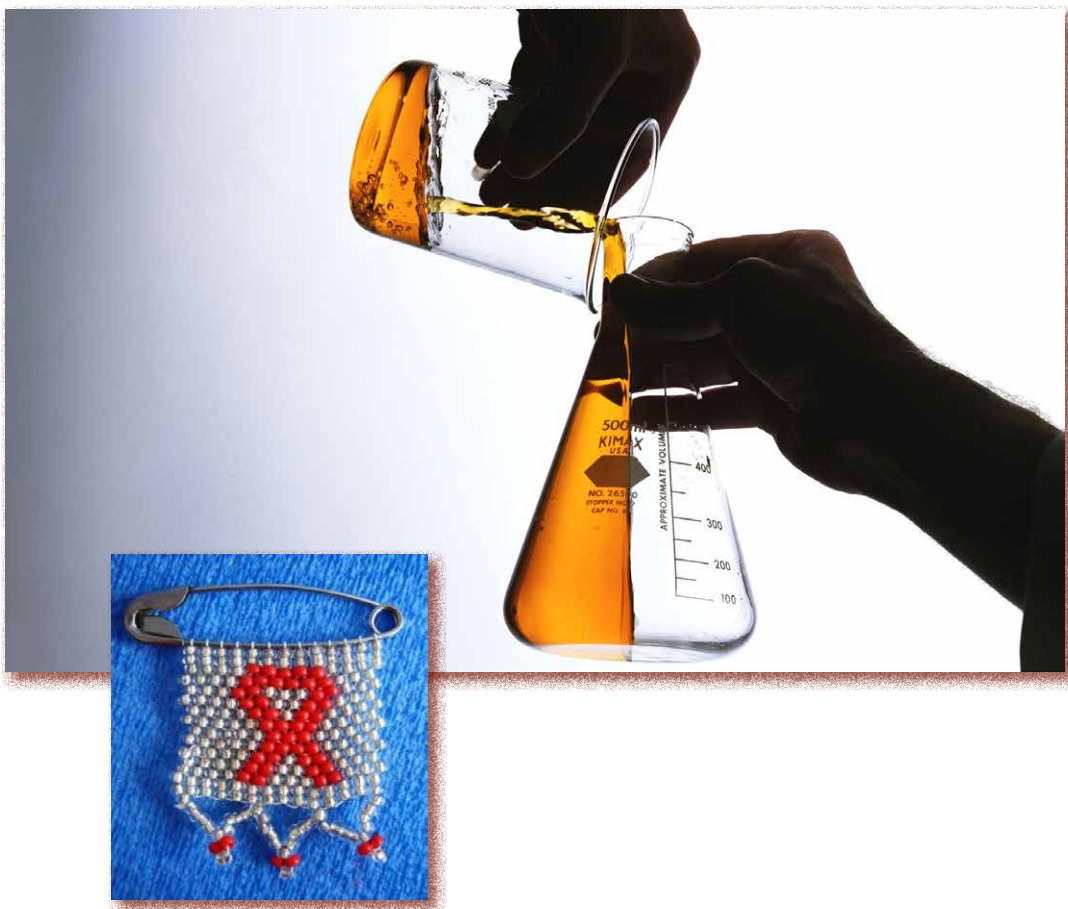


Table 3.7: Knowledge about HIV transmission among students and staff

Variable	Students		Staff		Test statistic
	n	%	n	%	
HIV can spread to males/females through unprotected sex					
True	5 388	96.3	950	96.6	$\chi^2 = 0.57$ p = 0.753
Don't know	73	1.3	10	1.0	
Sexually transmitted infections (STIs) put people at greater risk of HIV infection or infection with new forms of the virus					
True	4 602	83.8	869	89.7	$\chi^2 = 25.9$ p < 0.001
Don't know	698	12.7	68	7.0	
A person can have the AIDS virus and pass it to other people even if the person doesn't look sick					
True	4 588	83.9	865	89.0	$\chi^2 = 20.5$ p < 0.001
Don't know	383	7.0	34	3.5	
If both partners are HIV positive it is OK to have unprotected sex					
True	560	10.2	103	10.6	$\chi^2 = 34.4$ p < 0.001
Don't know	680	12.4	57	5.9	
Having anal sex increases your chances of getting infected with HIV					
True	2 599	48.0	590	61.4	$\chi^2 = 65.3$ p < 0.001
Don't know	2 054	37.9	244	25.4	
If a pregnant woman is infected with HIV, the virus can be passed to her unborn child					
True	3 878	71.3	792	82.5	$\chi^2 = 53.0$ p < 0.001
Don't know	618	11.4	75	7.8	
People can protect themselves from getting infected with HIV by not having sexual intercourse					
True	3 327	61.6	691	72.3	$\chi^2 = 47.4$ p < 0.001
Don't know	594	11.0	52	5.4	
AIDS can be cured					
True	878	16.2	115	12.0	$\chi^2 = 32.9$ p < 0.001
Don't know	758	14.0	87	9.1	
People can reduce their chances of getting the AIDS virus by using a condom every time they have sex					
True	4 932	89.7	895	92.7	$\chi^2 = 8.4$ p < 0.001
Don't know	214	3.9	30	3.1	

* Pearson chi square

3.2.3. Knowledge of and access to HIV prevention and treatment

Knowledge of where to get condoms was high: 98.3% of the respondents knew how and where to access condoms. There was no statistically significant difference in knowledge levels between students and staff, as 98.4% and 98.3% respectively provided positive answers (Table 3.8).

A slightly higher proportion of staff (93.9%) than students (91.4%) indicated that, if they wanted to, they could obtain a condom. Staff members were also more likely than students to say they were aware of medication that HIV-positive pregnant women can take to reduce the risk of infecting the baby (88.7% compared to 79.9%). Only 60% of students, compared with 75.1% of staff members, reported that they had heard about medicines to reduce the risk of HIV infection for women who had been exposed to HIV as a result of rape.

Table 3.8: Knowledge of and access to HIV prevention methods

Variable	Students		Staff		Test statistic
	n	%	n	%	
Know of a place where a person can get condoms					
Yes	5 485	98.4	959	98.3	$\chi^2= 3.1$ $p = 0.079$
No	87	1.6	23	1.7	
If you wanted to, could you get a condom					
Yes	5 086	91.4	923	93.9	$\chi^2 = 17.3$ $p < 0.001$
No	141	2.6	32	3.3	
Unsure	335	6.0	28	2.8	
Heard about medical treatment that HIV- positive pregnant women can take to reduce the risk of infecting the baby					
Yes	4 470	79.9	875	88.7	$\chi^2= 42.3$ $p < 0.001$
No	1 126	20.1	112	11.3	
Heard about medical treatments that can reduce the risk of HIV- infection if a woman has been raped					
Yes	3 346	60.0	741	75.1	$\chi^2 = 80.9$ $p < 0.001$
No	2 229	40.0	246	24.9	

* Pearson chi square

3.2.4. Knowledge of and exposure to TB

More than one-quarter of respondents (28.7%) reported lifetime testing for TB disease (Table 3.9). Overall, 20.5% of those participants who had ever been tested for TB were found to have the disease.

Less than a third of respondents (31.2%) said they had been exposed to someone with TB, while the majority (72.5%) said they had not been exposed.

Knowledge that TB can be cured was high among staff (92.7%) but significantly lower (85.9%) among students.

Table 3.9: Knowledge of and exposure to TB among students and staff

Variable	Students		Staff		Test statistic
	n	%	n	%	
Ever had a TB test					
Yes	1 544	28.1	317	32.5	$\chi^2 = 8.3$ $p = 0.016^*$
No	3 854	70.1	642	65.9	
Not sure	103	1.9	15	1.5	
If yes, were results positive for TB					
Yes	329	20.7	65	19.6	$\chi^2 = 5.3$ $p = 0.071$
No	1 143	71.8	252	76.1	
Not sure	121	7.6	14	4.2	
Ever been exposed to anyone with TB					
Yes	1 607	31.1	296	31.6	$\chi^2 = 0.9$ $p = 0.653$
No	3 088	59.8	564	60.3	
Not sure		9.0	76	8.1	
Can TB be cured					
Yes	4 339	85.9	905	92.7	$\chi^2 = 34.9$ $p < 0.001$
No	263	5.2	31	3.2	
Don't know	452	8.9	40	4.1	

* Pearson chi square

3.3. Attitudes to HIV prevention, care and treatment and related matters

3.3.1. Attitudes to HIV-positive people

As indicated in Table 3.10, 20.7% of students and 13.6% of staff said they did not want to be associated with HIV-positive people. The difference between students and staff on this variable was statistically significant ($p < 0.001$). In addition, more male students (16.1%) than female students (10.5%) did not want to be associated with HIV-positive people, and more white students (24.7%) than African (12.8%) students had this attitude.

A slightly higher proportion of staff (89.3%) than students (87.9%) said if a family member had HIV their relationship with the affected person would be good. Among students, potentially supportive attitudes for an infected family member increased with age, from 85.8% among students younger than 18 years, to 87.6% among those aged 18-24 years, and 91.3% among those aged 25 years and upward.

More students (23.8%) than staff members (19%) said that they would find it difficult to stay in the same house as HIV-positive people. This feeling was highest among students younger than 18 years (25.4%) and dropped to 23.8% among those aged 18-24 years, and to 20.8% among those aged 25 and older. More male students (24.8%) and male staff (21.5%) compared with female students (22.4%) and female staff (16.7%) stated they would find it difficult to share a house with HIV-positive people.

Table 3.10: Attitudes of students and staff to HIV-positive people

	I do not want to be with associated with HIV-positive people						If a family member has HIV my relationship with him/her will be good						Staying in the same house with HIV-positive people will be very difficult for me					
	Total		Agree		Disagree		Total		Agree		Disagree		Total		Agree		Disagree	
	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n	%
Total students Total staff	3 156 491	654 67	20.7 13.6	2 502 424	79.3 86.4		5 133 910	4 514 813	87.9 89.3	619 97	12.1 10.7	3 525 552	832 105	23.6 19	2 693 447	76.4 81		
Age																		
<18 students	337	82	24.3	255	75.7		487	418	85.8	69	14.2	319	81	25.4	238	74.6		
18-24 students	2 420	481	19.9	1 939	80.1		3 989	3 496	87.6	493	12.4	2 764	659	23.8	2 105	76.2		
25+ students	399	91	22.8	308	77.2		657	600	91.3	57	8.7	442	92	20.8	350	79.2		
Sex																		
Male students Male staff	1 644 234	375 36	22.8 15.4	1 269 198	77.2 84.6		2 524 417	2 174 363	86.1 87.1	350 54	13.9 12.9	1 782 256	442 55	24.8 21.5	1 340 201	75.2 78.5		
Female students Female staff	1 512 255	279 30	18.5 11.8	1 233 225	81.5 88.2		2 609 491	2 340 448	89.7 91.2	269 43	10.3 8.8	1 743 294	390 49	22.4 16.7	1 353 245	77.6 83.3		
Race																		
African Students Staff	2 722 337	559 48	20.5 14.2	2 163 289	79.5 85.8		4 550 688	4 032 623	88.6 90.6	518 65	11.4 9.4	3 038 381	689 71	22.7 18.6	2 349 310	77.3 81.4		
White Students Staff	200 76	54 8	27 10.5	146 68	73 89.5		255 111	212 93	83.1 83.8	43 18	16.9 16.2	213 89	80 18	37.6 20.2	133 71	62.4 79.8		
Coloured Students Staff	199 63	35 8	17.6 12.7	164 55	82.4 87.3		274 91	222 78	81 85.7	52 13	19 14.3	231 66	57 12	24.7 18.2	174 54	75.3 81.8		
Indian/Asian Students Staff	31 10	6 1	19.4 10	25 9	80.6 90		46 14	41 13	89.1 92.9	5 1	10.9 7.1	38 11	5 2	13.2 18.2	33 9	86.8 81.8		

Data presented in Table 3.11 show that a higher proportion of students (19.9%) than staff (14.2%) took the view that people who become infected with HIV are promiscuous. More male students (17.1%) than female students (11.6%) subscribed to this view.

In addition, a higher proportion of male students (21.8%) than female students (17.3%) said that HIV-negative people should not be allowed to socialise with HIV- positive people. This negative view was most widely held among students in the Free State (37.1%) and least prevalent among students in the Eastern Cape (6.8%).



Table 3.11: Attitudes of students and staff to HIV-positive people

Variables	People who get infected with HIV are promiscuous						HIV-negative people should not be allowed to socialise with HIV-positive people					
	Total	Agree		Disagree		Total	Agree		Disagree			
	n	n	%	n	%	n	n	%	n	%		
Total students Total staff	4 115 614	820 87	19.9 14.2	3 295 527	80.1 85.8	2 373 407	511 79	21.5 19.4	1 862 328	78.5 80.6		
Age												
<18	384	79	20.6	305	79.4	255	60	23.5	195	76.5		
18-24 students 18-24 staff	3 238 68	651 7	20.1 10.3	2 587 61	79.9 89.7	1 808 24	376 5	20.8 20.8	1 432 19	79.2 79.2		
25+ students 25+ staff	493 546	90 80	18.3 14.7	403 466	81.7 85.3	310 382	75 74	24.2 19.3	235 309	75.8 80.7		
Sex												
Male Students Staff	2 114 292	449 50	21.2 17.1	1 665 242	78.8 82.9	1 256 197	297 43	21.7 21.8	1 585 154	78.3 78.2		
Female Students Staff	2 001 320	371 37	18.5 11.6	1 630 283	81.5 88.4	1 117 208	214 36	19.2 17.3	903 172	80.8 82.7		
Race												
African Students Staff	3 602 446	721 62	20 13.9	2 881 384	80 86.1	2 025 269	440 55	21.7 20.4	1 585 214	78.3 79.6		
White Students Staff	225 86	57 14	25.3 16.3	168 72	74.7 83.7	163 71	34 11	20.9 15.5	129 60	79.1 84.5		
Coloured Students Staff	246 66	34 8	13.8 12.1	212 58	86.2 87.9	155 57	30 9	19.4 15.8	125 48	79.2 84.2		
Indian Students Staff	37 11	7 2	18.9 18.2	30 9	81.1 81.8	24 7	5 2	20.8 28.6	19 5	79.2 71.4		

3.3.2. Attitudes to treatment for HIV and AIDS

Students and staff at TVET colleges were generally positive about treatment for HIV and AIDS.

As indicated in Table 3.12, 78.8% of students and 77.3% of staff said they trusted that HIV treatment worked. However, 24.3% of students younger than 18 years, 20.9% of those aged 18-24 years and 20.8% of students older than 25 years said they did not trust that HIV treatment worked. Mistrust about HIV treatment was highest among students in the Western Cape (24.5%) and lowest in Mpumalanga (9.4%).

Most respondents – 80.4% of students and 82.4% of staff – agreed that HIV treatment would keep an HIV-positive person alive. This view was held by 77.7% of students younger than 18 years, 79.5% of those aged 18-24 years and 88% of those 25 years and older. It was most strongly supported among female staff (83.3%) and least supported among male students (78.3%). In geographic terms, the most positive student attitudes to HIV treatment were in the Eastern Cape (91.9%) and North West (91.2%) and the least positive attitudes were in the Western Cape (83.6%).

More than a quarter of students and staff (27.8%) incorrectly said that you could share HIV treatment with your partner. However, a strong majority of students and staff (72.2%) were correctly informed that it was not right to share HIV treatment with your partner.



Table 3.12: Attitudes of students and staff to treatment for HIV and AIDS

Variables	I don't trust that HIV treatment works				HIV treatment will keep an HIV-positive person alive				You can share HIV treatment with your partner						
	Total	Agree		Disagree		Total	Agree		Disagree		Total	Agree		Disagree	
	N	n	%	n	%	n	%	n	%	n	%	n	%	n	%
Total students Total staff	3 721 616	788 140	21.2 22.7	2 933 476	78.8 77.3	5 191 898	4 176 740	80.4 82.4	1 015 158	19.6 17.6	3 074 474	855 132	27.8 27.8	2 219 342	72.2 72.2
Age															
<18 students	342	83	24.3	259	75.7	484	376	77.7	108	22.3	297	85	28.6	212	71.4
18-24 students 18-24 staff	2 898 59	605 13	20.9 22	2 293 46	79.1 78	4 039 96	3 212 76	79.5 79.2	827 20	20.5 20.8	2 410 47	681 10	28.3	1 729	71.7 78.7
25+ students 25+ staff	481 557	100 127	20.8 22.8	381 430	79.2 77.2	668 802	588 664	88.0 82.2	80 138	12 17.2	367 427	89 122	24.3 21.3	278 37	75.7 71.4
Sex															
Male Students Staff	1 898 285	449 77	23.7 27	1 449 208	76.3 73	2 562 410	2 007 334	78.3 81.5	555 76	21.7 18.5	1 550 225	450 65	29 28.9	1 100 160	71 71.1
Female Students Staff	1 823 329	339 63	18.6 19.1	1 484 266	81.4 80.9	2 629 486	2 169 405	82.5 83.3	460 81	17.5 16.7	1 524 247	405 66	26.6 26.7	1 119 181	73.4 73.3
Race															
African Students Staff	3 222 435	682 101	21.2 23.2	2 540 334	78.8 76.8	4 597 678	3 772 579	82.1 85.4	825 99	17.9 14.6	2 700 354	767 104	28.4 29.4	1 933 250	71.6 70.6
White Students Staff	224 89	50 15	22.3 16.9	174 74	77.7 83.1	251 109	172 73	68.5 67	79 36	31.5 33	169 65	41 20	24.3 30.8	128 45	75.7 69.2
Coloured Students Staff	233 76	49 21	21 27.6	184 55	79 72.4	285 92	182 71	63.9 77.2	103 21	36.1 22.8	168 45	33 6	19.6 13.3	135 39	80.4 86.7
Indian/Asian Students Staff	36 11	6 2	16.7 18.2	30 9	83.3 81.8	50 14	44 13	88 92.9	6 1	12 7.1	32 6	13 1	40.6 16.7	19 5	59.4 83.3

Students and staff at TVET colleges had positive attitudes towards antiretroviral medicines and their effectiveness in treating HIV. Overall, 83.2% of respondents disagreed that antiretroviral drugs were poisonous (Table 3.13) while 16.8% regarded these medicines as poisonous. The view that ARVs were poisonous was held more strongly among students (19.8%) than staff (13.8%), with quite a large difference between male students (22.6%) and male staff (14.7%) and a slightly smaller difference between female students (16.8%) and female staff (12.9%). The most negative attitudes to HIV medication were in the Northern Cape and Limpopo (where 16.2% of respondents viewed it as poisonous) and negative attitudes to HIV medication were least often expressed in the Eastern Cape (8.2%).

Most respondents (73.3%) agreed that HIV medication really works. This belief in the efficacy of ARVs was held by 72.4% of students and 74.2% of staff. Female students and female staff members were more likely to believe that ARVs really worked (75.9% and 77.2% respectively held this view) than male students (68.7%) and male staff (70.6%). The belief that ARVs really work was most strongly held among students in Gauteng (88.3%) and Mpumalanga (87.9%) and least common in the North West (81.3%).



Table 3.13: Attitudes of students and staff to treatment for HIV and AIDS

Variables	ARVs (antiretroviral or HIV medicine) are poisonous				HIV medicine really works			
	Agree		Disagree		Agree		Disagree	
	Total n	n	%	n	%	n	n	%
Total students Total staff	3 080 455	609 63	19.8 13.8	2 471 392	80.2 86.2	3 583 631	1 368 220	27.6 25.9
Age								
Students (15+)	3 080	609	19.8	2 471	80.2	3 583	1 368	27.6
Staff (18+)	457	63	13.8	394	86.2	633	220	25.8
Sex								
Male Students Staff	1 586 231	358 34	22.6 14.7	1 228 197	77.4 85.3	1 663 276	757 115	31.3 29.4
Female Students Staff	1 494 224	251 29	16.8 12.9	1 243 195	83.2 87.1	1 920 355	611 105	24.1 22.8
Race								
African Students Staff	2 696 330	549 44	20.4 13.3	2 147 286	79.6 86.7	3 283 521	1 115 126	25.4 19.5
White Students Staff	166 65	25 9	15.1 13.8	141 56	84.9 86.2	119 55	116 46	49.4 45.5
Coloured Students Staff	184 51	25 7	13.6 13.7	159 44	86.4 86.3	145 43	119 43	45.1 50
Indian/Asian Students Staff	28 6	9 1	32.1 16.7	19 5	67.9 83.3	33 10	14 3	29.8 23.1

3.3.3. Attitudes to condom use

Both students and staff at TVET colleges held negative attitudes to condom use.

As indicated in Table 3.14, 40.8% of students and 43% of staff who responded to the question said condoms felt unnatural. This opinion was most widely held among students in the Western Cape (46.6%) and least prevalent among students in the Northern Cape (33.3%).

Slightly more staff (31.3%) than students (29.8%) said condoms changed the climax.

About one in five students and staff took the view that a woman loses a man's respect if she asks him to use a condom. This view was slightly more prevalent among students (21.9%) than staff (20%). Significantly more Indian students (29.7%) than Indian staff (10%) concurred with this opinion.



Table 3.14: Attitudes of students and staff to condom use

Variables	Condoms feel unnatural						Condoms change the climax						A woman loses a man's respect if she asks him to use a condom					
	Agree			Disagree			Agree			Disagree			Agree			Disagree		
	Total	n	%	n	%	Total	n	%	n	%	Total	n	%	n	%	Total	n	%
Total students Total staff	4 392 777	1 790 334	40.8 43	2 602 443	59.2 57	4 377 769	1 304 240	29.8 31.2	3 073 529	70.2 68.8	3 700 643	810 129	21.9 20.1	2 890 514	78.1 79.9			
Age in years																		
<18 students	437	160	36.6	277	63.4	432	134	31	298	69	363	78	21.5	285	78.5			
18-24 Students Staff	3 393 72	1 391 35	41 48.6	2 002 37	59 51.4	3 394 72	1 004 22	29.6 30.6	2 390 50	70.4 69.4	2 853 55	615 12	21.6 21.8	2 238 43	78.4 78.2			
25+ Students Staff	562 706	239 299	42.5 42.4	323 407	57.5 57.6	551 699	166 219	30.1 31.3	385 480	69.9 68.7	484 590	117 117	24.2 19.8	367 473	75.8 80.2			
Sex																		
Male Students Staff	2 270 382	1 164 191	51.3 50	1 106 191	48.7 50	2 264 377	853 152	37.7 40.3	1 411 225	62.3 59.7	1 948 322	488 76	25.1 23.6	1 460 246	74.9 76.4			
Female Students Staff	2 122 395	626 143	29.5 36.2	1496 252	70.5 63.8	2 113 392	451 88	21.3 22.4	1 662 304	78.7 77.6	1 752 321	322 53	18.4 16.5	1 430 268	81.6 83.5			
Race																		
African Students Staff	3 876 585	1 598 249	41.2 42.6	2 278 336	58.8 57.4	3 863 577	1 141 184	29.5 31.9	2 722 393	70.5 68.1	3 256 479	737 100	22.6 20.9	2 519 379	77.4 79.1			
White Students Staff	224 93	97 43	43.3 46.2	127 50	56.7 53.8	228 95	68 27	29.8 28.4	160 68	70.2 71.6	182 80	39 16	21.4 20	143 64	78.6 80			
Coloured Students Staff	244 83	78 38	32 45.8	166 45	68 54.2	237 82	80 27	33.8 32.9	157 55	66.2 67.1	218 73	21 12	9.6 16.4	197 61	90.4 83.6			
Indian Students Staff	41 13	15 4	36.6 30.8	26 9	63.4 69.2	42 13	13 2	31 15.4	29 11	69 84.6	37 10	11 1	29.7 10	26 9	70.3 90			

Overall, more than a quarter of students (26%) and staff (26.9%) said they used condoms only if their sexual partners wanted them to. The Eastern Cape had the highest rate of students reporting non-use of condoms unless their partners insisted (30.1%) while the North West had the lowest rate (15.2%).

Slightly more than one-third of students (34.2%) and nearly a third of staff (32.9%) said condoms should only be used if having sex with a person who is not the main partner. More male students (40.1%) and staff (39.6%) than female students (27.3%) and staff (25.3%) took this view. This attitude was most frequently held among students in the Eastern Cape (29.8%) and least prevalent among students in the North West (15.9%).

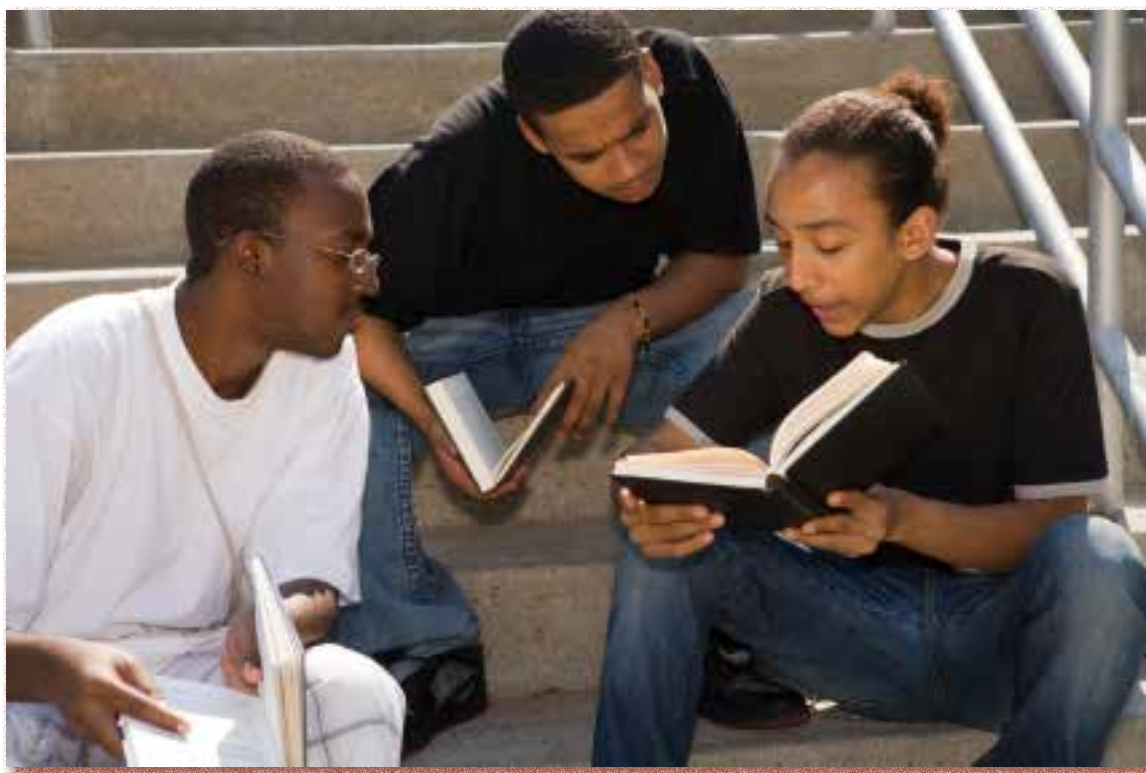


Table 3.15: Attitudes of students and staff to condom use

Variables	I use condoms only if my sexual partner wants us to use them				Condoms should only be used if having sex with a person who is not the main partner					
	Agree		Disagree		Total	Agree		Disagree		
	n	%	n	%	n	n	%	n	%	
Total students Total staff	3 717 665	968 179	26 26.9	2 749 486	74 73.1	3 286 599	1 124 197	34.2 32.9	2 162 402	65.8 67.1
Age in years										
<18 students	353	96	27.2	257	72.8	328	124	37.8	204	62.2
18-24 Students Staff	2 882 57	731 14	25.4 24.6	2 151 43	74.6 75.4	2 507 45	821 12	32.7 26.7	1 686 33	67.3 73.3
25+ Students Staff	482 608	141 165	29.3 27.1	341 443	70.7 72.9	451 554	179 185	39.7 33.4	272 369	60.3 66.6
Sex										
Male students Male staff	1 984 325	612 108	30.8 33.2	1 372 217	69.2 66.8	1 782 308	714 122	40.1 39.6	1 068 186	59.9 60.4
Female students Female staff	1 733 338	356 71	20.5 21	1 377 267	79.5 79	1 504 289	410 73	27.3 25.3	1 094 216	72.7 74.7
Race										
African Students Staff	3 253 495	865 129	26.6 26.1	2 388 366	73.4 73.9	2 884 445	986 142	34.2 31.9	1 898 303	65.8 68.1
White Students Staff	204 81	51 26	25 32.1	153 55	75 67.9	165 79	57 25	34.5 31.6	108 54	65.5 68.4
Coloured Students Staff	219 73	43 22	19.6 30.1	176 51	80.4 69.9	201 62	75 27	37.3 43.5	126 35	62.7 56.5
Indian/Asian Students Staff	36 12	7 2	19.4 16.7	29 10	80.6 83.3	32 11	4 3	12.5 27.3	28 8	87.5 72.7

3.3.4. Attitudes to partner testing for STIs

Attitudes of students and staff to the prevention of STIs revealed a high degree of uncertainty. About four out of 10 respondents said they did not know what their partner's response would be if they were to ask him or her to get tested for STIs.

More than one-third agreed or strongly agreed that a request to the partner to get tested for STIs would make the partner suspect that the respondent had been cheating or was accusing the partner of infecting him/her. Only 21.7% disagreed or strongly disagreed that their partner would suspect them of cheating, while 23.8% were confident their partner would not think he/she was being accused of infecting the respondent.

Table 3.16: Attitudes to partner testing for STIs among students and staff

Variable	Strongly agree		Agree		Don't know		Disagree		Strongly disagree	
	n	%	n	%	n	%	n	%	n	%
If I ask my main sexual partner to get tested for STIs he may think that I cheated on him	1 033	16.3	1 321	20.8	2 613	41.2	569	9	802	12.7
If I ask my main sexual partner to get tested for STIs he may think that I believe that he gave me an STI	935	14.9	1 222	19.4	2 639	42	620	9.9	873	13.9

3.3.5. Attitudes to LGBTI people

The composition of the sample in terms of sexual orientation was as follows: 83.1% of students and 87.4% of staff identified themselves as heterosexual; 7.3% of students and 5.3% of staff identified themselves as homosexual; and 5.2% of students and 3.6% of staff as bisexual.

In terms of sexual preferences, anal sex accounted for 10.8% of student and 7.5% of staff responses, and oral sex for about 23% of student and 20.7% of staff selections. Vaginal sex was indicated in 71.7% of student answers and 67.5% of staff responses.

Attitudes expressed through the survey to lesbian, gay, bisexual, transgender and intersex (LGBTI) individuals were generally positive.

More than half of students (56.7%) and 70.2% of staff said they felt safe from physical harm on campus. Asked about the safety of the college environment for LGBTI individuals, 53.4% of students and 61% of staff said they considered their institution a safe environment for LGBTI students, while 60.3% of students and 65.7% of staff said they considered their institution an accepting environment for LGBTI students.

However, only 21.9% of students and 41.8% of staff said the management of the institution took discrimination against the LGBTI community seriously.

Furthermore, only 6.6% of students and 10.2% of staff said they had attended a meeting or function about LGBTI issues on campus while 9.4% of students and 12.3% of staff said they had received information about LGBTI issues on campus.

Table 3.17: Attitudes of students and staff to LGBTI people

Variables	Students		Staff		Total
	n	%	n	%	n
Sexual orientation					
Homosexual	384	7.3	47	5.3	431
Bisexual	277	5.2	32	3.6	309
Heterosexual	4 399	83.1	780	87.4	5 179
Other	232	4.4	33	3.7	265
Preferred form of sex *					
Anal sex	611	10.8	91	7.5	702
Oral sex	1 300	23.0	251	20.7	1 551
Vaginal sex	4 053	71.7	819	67.5	4 872
None	903	15.9	53	4.4	956
LGBTI questions					
Do you consider your institution a safe environment for LGBTI students					
Yes	2 627	53.4	491	61	3 118
No	511	10.4	70	8.7	581
Not sure	1 782	36.2	244	30.3	2 026
Do you consider your institution an accepting environment for LGBTI students					
Yes	3 154	60.3	583	65.7	3 737
No	595	11.4	85	9.6	680
Not sure	1 482	28.3	220	24.8	1 702
I feel safe from physical harm on campus					
Yes	2 957	56.7	620	70.2	3 577
No	863	16.5	101	11.4	964
Not sure	1 396	26.8	162	18.3	1 558
The management of the institution takes discrimination against LGBTI community seriously					
Yes	1 111	21.9	361	41.8	1472
No	1 528	30.1	203	23.5	1 731
Not sure	2 442	48.1	299	34.6	2 741
I attended a meeting or function about LGBTI issues on campus					
Yes	306	6.6	82	10.2	388
No	3 650	78.1	611	76.3	4 261
Not sure	715	15.3	108	13.5	823
I have received information about LGBTI issues on campus					
Yes	432	9.4	98	12.3	530
No	3 479	75.7	602	75.5	4 081
Not sure	686	14.9	97	12.2	783

*Some respondents selected more than one answer

3.4. Behaviours that put students and staff at risk of HIV infection

3.4.1. Sexual relationships of students and staff

A total of 60.9% of students and 72% of staff at TVET colleges indicated that they had a current sexual partner (Table 3.18). Among students significantly more males than females indicated that they had a sexual partner, while the gap between staff members of different sexes was marginal.

About half of male and female students who were in relationships had a partner between two and four years older or younger than themselves. In line with social convention, male students tended to have partners younger than themselves and for female students the reverse situation existed.

In the case of 24.2% of students who were in relationships there was an age gap of five or more years between the respondent and his/her partner. Female students comprised 55.6% of the group with age-disparate partnerships and in eight out of 10 of these relationships the partner was older. Male students in age-disparate relationships had a partner younger than themselves in six out of 10 cases.

Male students were significantly more likely to report being in a relationship with someone of a similar age when compared to female students.

Among staff members (who were substantially older than students) there was a stronger tendency (compared to students) to have partners somewhat older or younger than themselves. Only 13.2% of staff members in relationships had a partner the same age as themselves.

Approximately 48% of students and 46% of staff said they had only had a male sexual partner in the past year, while 51% of students and 53% of staff said they had only had a female a sexual partner in the past year. A significant proportion of males reported having a women partner and the opposite was true for women, for both students and staff. As indicated above, the great majority of these relationships were heterosexual. Overall a very small proportion of respondents were bisexual and only 0.05% of students and 0.03% of staff said they had had both female and male sexual partners in the past year, and there was no difference by sex for both students and staff.



Table 3.18: Sexual relationships of students and staff at TVET colleges

Variables	Students						Staff					
	Male			Female			Male			Female		
	Total	n	%	n	%	p-value	Total	n	%	n	%	p-value
Do you have a sexual partner?												
Yes	3 251	1 643	63.1	1 608	59	0.002	665	325	49	340	51	0.005
No	2 085	962	36.9	1 123	41		259	100	39	159	61	
Partner five or more years older/younger												
Younger	278	202	60.3	76	18	<0.001	125	105	84	20	16	<0.001
Older	478	133	39.7	345	82		145	28	19	117	81	
Partner between two and four years older/younger												
Younger	700	579	75.5	121	17	<0.001	119	98	82	21	18	<0.001
Older	768	188	24.5	580	83		106	25	24	81	76	
Partner same age as you	723	454	62.8	269	37	0.004	75	41	55	34	45	0.944
Sexual partners in the past year												
Only men Yes No												
	1 931	154	11	1 777	72	<0.001	401	40	10	361	90	<0.001
Only women Yes No												
	1 920	1 237	88.7	683	28		272	195	72	77	28	
Both men and women Yes No												
	1 841	1 727	73.3	114	8.9	<0.001	328	315	96	13	4	<0.001
	1 790	629	26.7	1 161	91		281	55	20	226	80	
	170	96	7.3	74	5.9	0.157	21	13	62	8	38	0.194
	2 391	1 216	92.7	1 175	94		422	200	47	222	53	

3.4.2. Condom use among students and staff at TVET colleges

The extent of condom use at last sex by students and staff at TVET colleges varied according to type of partner.

As indicated in Table 3.19, 55.1% of students and 34.5% of staff said they had used a condom the last time they had sex with their partner. The difference between students and staff in this respect was statistically significant at $p < 0.001$.

More than half the students (56.5%) and 51.5% of staff said they used a condom “every time” with a regular partner. A further 23.3% of students and 20.3% of staff said they used condoms “almost every time”. Only 16.8% of students and 21.6% of staff said they use a condom “sometimes”, and 1.8% of students and 3.6% of staff said that they never used a condom with a partner.

Most students (75.5%) and staff (76.3%) said they used condoms the last time they had sex with non-regular partners. More than half (57.7%) of students and (54.8%) of staff said they used condoms with non-regular partner “every time”, 15.9% of students and 18.8% of staff said “almost every time”, 10.8% of students and 12.9% of staff said “sometimes” and 5.9% of students and 8.6% of staff said they never used a condom with a non-regular partner.

Only half of students (51%) and 33% of staff said they had used a condom the last time they had sex with commercial sex worker during, while 49% of students and 67% of staff said they did not use a condom during their last sexual act with a commercial sex worker.

Slightly more students (61%) than staff (59%) said they “always” used a condom with a commercial sex worker, while 21% and 15% of students and staff respectively said they “almost always” used a condom in this situation.



Table 3.19: Condom use among students and staff at TVET colleges

Variables	Student		Staff		Test statistic
	n	%	n	%	
Condom use at last sex with partner					
Yes	2 761	55.1	296	34.5	$\chi^2 = 175.4$ p < 0.001*
No	1 750	34.9	504	58.7	
Consistent condom use with partner					
Every time	1 583	56.5	157	51.5	$\chi^2 = 13.7$ p = 0.008
Almost every time	653	23.3	62	20.3	
Sometimes	470	16.8	66	21.6	
Never	51	1.8	11	3.6	
Don't know	43	1.5	9	3.0	
Condom use at last sex with non-regular partner					
Yes	1 647	75.5	145	76.3	$\chi^2 = 6.2$ p = 0.045
No	285	13.1	33	17.4	
Don't know	243	11.2	12	6.3	
Consistent condom use with non-regular partner					
Every time	1 204	57.7	102	54.8	$\chi^2 = 8.2$ p = 0.085
Almost every time	331	15.9	35	18.8	
Sometimes	226	10.8	24	12.9	
Never	124	5.9	16	8.6	
Don't know	203	9.7	9	4.8	
Condom use at last sex with commercial sex worker					
Yes	1 308	51	123	33	$\chi^2 = 44.4$ p < 0.001
No	1 244	49	252	67	
Consistent condom use with commercial sex worker					
Every time	896	61	92	59	$\chi^2 = 17.0$ p = 0.002
Almost every time	305	21	24	15	
Sometimes	188	13	22	14	
Never	72	5	18	12	

* Pearson chi square

3.4.3. Circumcision status of male students and staff at TVET colleges

The rate of male circumcision, as self-reported by male students and staff, was high. There was also strong support for medical male circumcision (MMC) among students and staff at TVET colleges.

As indicated in Table 3.20, almost two-thirds of male students (65.5%) and more than half of male staff (53.9%) said that they had been circumcised. Hospital-based circumcision was preferred by 76.4% of students and 84.6% of staff. The “bush”, meaning traditional circumcision,

was supported by 17.6% of students and 11.5% of staff, while 4.1% of students and 2.2% of staff preferred circumcision to be conducted at home.

There was overwhelming support for medical male circumcision (MMC), with 85.7% of staff and 78.5% of students preferring medical circumcision and only 21.5% of students and 14.3% of staff choosing traditional circumcision.

When asked who should perform circumcision, 87.8% of staff and 81% of students said a doctor, 15.4% of students and 10.4% of staff said a traditional healer or traditional surgeon, and only 1.6% of students and 1% of staff said a faith healer or church pastor.

Table 3.20: Male circumcision status and attitudes to circumcision

Variables	Male students		Male staff		Total
	n	%	n	%	n
Have you ever been circumcised?					
Yes	2 011	65.5	262	53.9	2 273
No	1 057	34.5	224	46.1	1 281
Where should circumcision be done?					
Home	130	4.1	12	2.2	142
Hospital	2 411	76.4	471	84.6	2 882
In the bush	556	17.6	64	11.5	620
Other	58	1.8	10	1.8	68
What kind of circumcision would you prefer for yourself or your partner?					
Medical	2 567	78.5	497	85.7	827
Traditional	703	21.5	83	14.3	1 478
Who should perform circumcision?					
Doctor	2 639	81	502	87.8	3 141
Faith healer/church pastor	53	1.6	6	1	59
Traditional healer/surgeon	502	15.4	61	10.7	563
Other	63	1.9	3	0.5	66

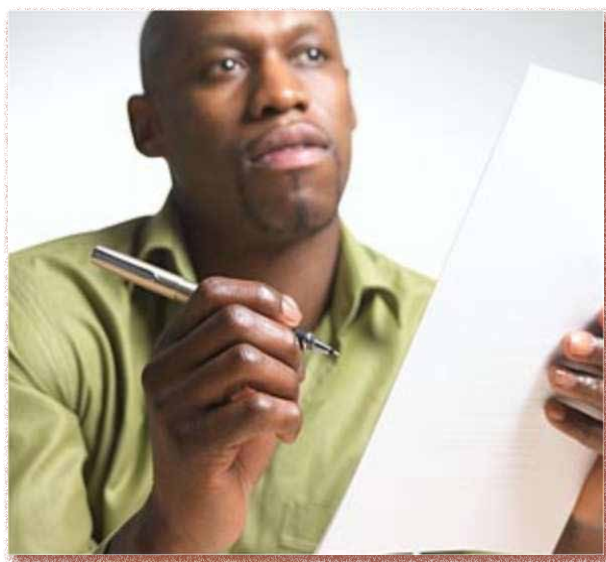


Figure 3.2 shows self-reported circumcision status according to race group. Circumcision was highest among Africans (70.0%), followed by Indians (55.6%), coloureds (47.2%) and whites (36.8%).

Figure 3.2: Male circumcision status by race group

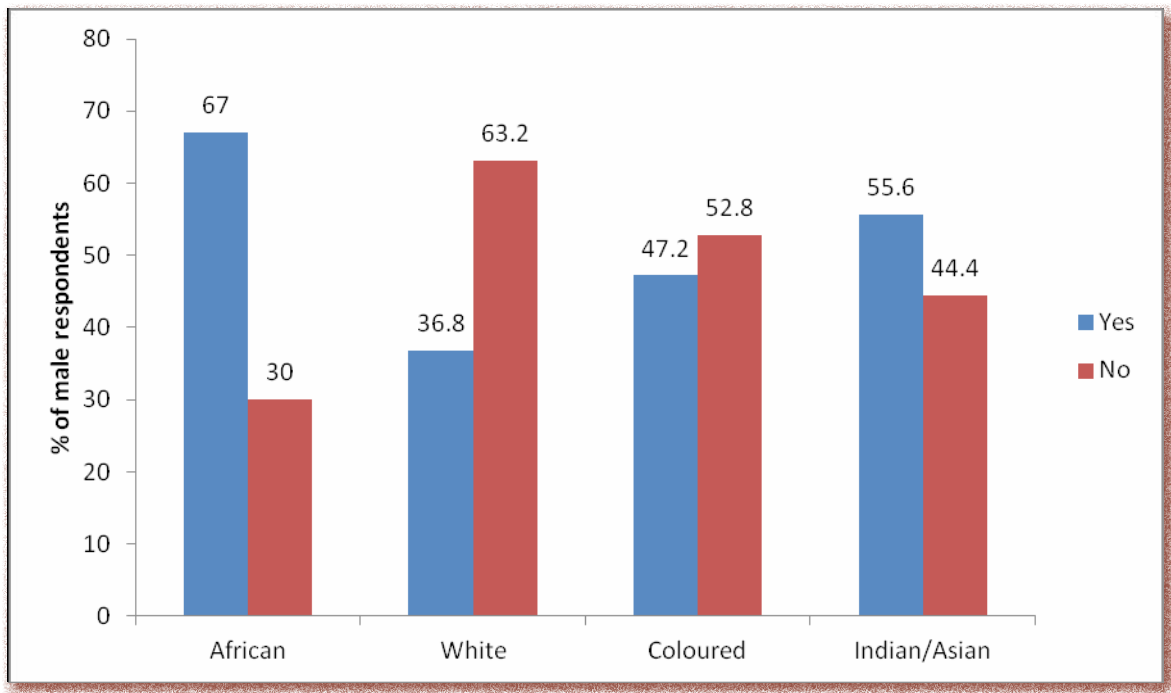
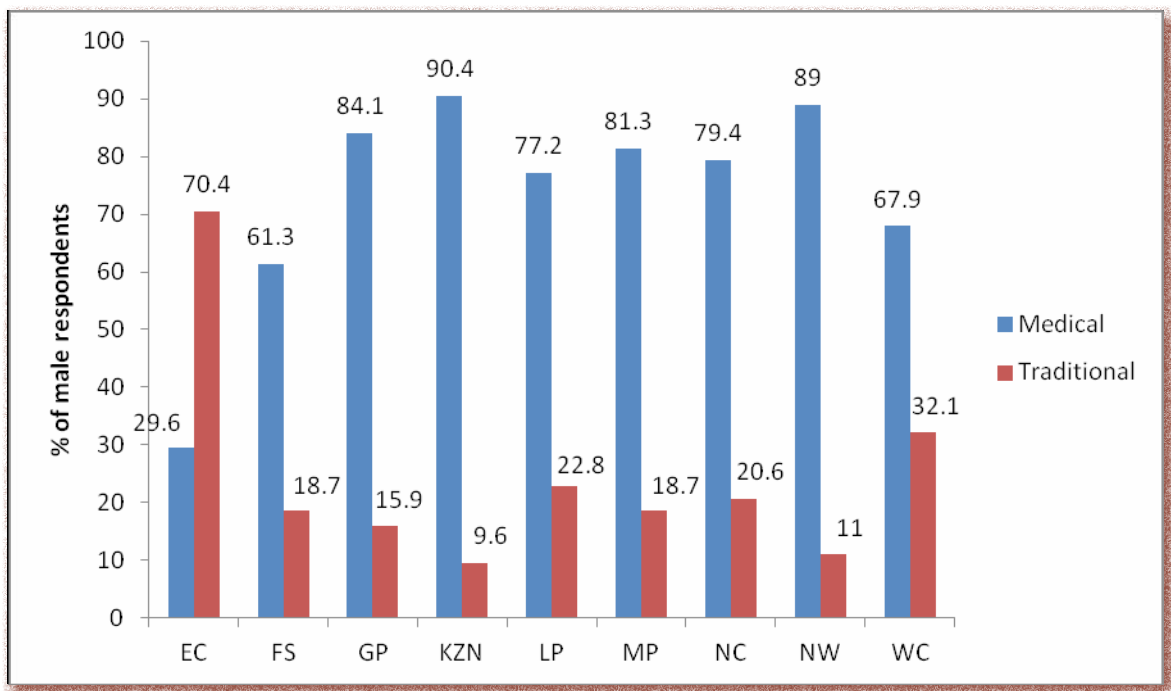


Figure 3.3 shows the preferred methods of circumcision by province. Medical circumcision was clearly selected over traditional circumcision in every province except the Eastern Cape. MMC was most strongly favoured in KwaZulu-Natal (90.4%), North West (89.0%) and Gauteng (84.1%) and received its lowest rate of support in the Eastern Cape (29.6%). Traditional circumcision was strongly supported in the Eastern Cape (70.4%) and received its next strongest approval in the Western Cape (32.1%).

Figure 3.3: Preferred form of circumcision by province



3.4.4. Substance use among students at TVET colleges

This section explores the use of substances such as alcohol, drugs and tobacco by students and staff at TVET colleges.

About 59.5% of student respondents and 57.5% of staff members indicated that they consumed alcohol. As shown in Table 3.21, 8.7% of students who consumed alcohol were younger than 18 years and 91.3% were 18 years or older. The composition of the group of students who consumed alcohol was: 87% African, 5.8% White, 6.4% Coloured and 0.8% of Indian origin. The composition of the group of staff members who consumed alcohol was 71% African, 16.6% white, 10.8% coloured and 1.6% of Indian origin.

For students, the rates of alcohol use were roughly in proportion to the representation of the various racial groups in the TVET student community, with a slightly elevated alcohol use rate among white and coloured students. Among staff members the relative over-representation of white and coloured respondents in the sub-group who use alcohol was more pronounced.

The provincial share of student alcohol use is reflected in Table 3.19. Gauteng (24.9%), KwaZulu-Natal (15.6%) and Free State (12.7%) accounted for more than half the total, while North West, Mpumalanga (both 6.9%) and the Eastern Cape (5.2%) were at the low end of the scale. Among staff, KwaZulu-Natal (18.7%), Gauteng (15.3%) and Free State (13.5%) again represented a large share, with North West (8.1%) and the Eastern Cape (5.5%) accounting for a modest share.

Table 3.21: Alcohol use among students and staff by age, race and province

Variables	Student		Staff		Total
	n	%	n	%	n
Age					
Less than 18 years	293	8.7	0	0	293
Greater than or equal to 18 years	3 075	91.3	577	100	3 652
Race					
Black African	2 925	87	406	71	3 331
White	196	5.8	95	16.6	291
Coloured	215	6.4	62	10.8	277
Indian/Asian	26	0.8	9	1.6	35
Province					
Western Cape	424	12.6	69	12	493
Eastern Cape	176	5.2	32	5.5	208
Northern Cape	236	7	37	6.4	273
Free State	427	12.7	78	13.5	505
KwaZulu-Natal	525	15.6	108	18.7	633
North West	233	6.9	47	8.1	280
Gauteng	838	24.9	88	15.3	926
Mpumalanga	231	6.9	64	11.1	295
Limpopo	278	8.3	54	9.4	332
Total	3 368	100	577	100	3 945

Substance use among students and staff at TVET colleges was very low except for marijuana or dagga. About one in 10 students (11%) and 5% of staff said they had used dagga/marijuana. Only 1.1% of students and 0.6% of staff said they had used mandrax², 1% of students and 0.6% of staff said they had used tik³, 1.2% of students and 0.7% of staff said they had used nyaope⁴, 0.9% of students and 0.6% of staff said they had used wonga⁵, 1.5% of students and 0.7% of staff said they had used cocaine/crack or ecstasy, and 0.9% of students and 0.6% of staff said they had used inhalants like glue.

Table 3.22: Substance use among students and staff at TVET colleges

	Students		Staff		
Have you ever used:	n	%	n	%	
Dagga/marijuana					
Yes	462	11	32	5	$\chi^2 = 29.6$ $p < 0.001^*$
No	3 682	89	674	95	
Mandrax					
Yes	43	1.1	4	0.6	$\chi^2 = 1.9$ $p = 0.384$
No	4 021	98.9	689	99.4	
Tik					
Yes	41	1	4	0.6	$\chi^2 = 1.7$ $p = 0.428$
No	4 012	99	688	99.4	
Nyaope					
Yes	50	1.2	5	0.7	$\chi^2 = 1.9$ $p = 0.393$
No	4 002	98.8	688	99.3	
Wonga					
Yes	36	0.9	4	0.6	$\chi^2 = 0.7$ $p = 0.399$
No	3 980	99.1	688	99.4	
Cocaine/crack					
Yes	59	1.5	5	0.7	$\chi^2 = 2.5$ $p = 0.117$
No	3 961	98.5	688	99.3	
Inhalants like glue					
Yes	37	0.9	4	0.6	$\chi^2 = 0.8$ $p = 0.371$
No	3 979	99.1	687	99.4	
Ecstasy					
Yes	59	1.5	5	0.7	$\chi^2 = 2.4$ $p = 0.116$
No	3 945	98.5	686	99.3	

* Pearson chi square

Table 3.23 outlines tobacco use among students and staff at TVET colleges. Cigarettes were used by 21.5% of students and 18% of staff, snuff by 13.3% of students and 2.1% of staff, and chewing tobacco was used by 7.2% of students and 0.8% of staff.

² Mandrax was initially marketed as a sedative or sleeping tablet by French pharmaceutical giant Roussell Laboratories. It turned out to be highly addictive and was banned in 1977.

³ Tik is the slang name given to the stimulant drug, methamphetamine. Tik and all other forms of crystal meth are stimulants. They increase activity between the brain and the central nervous system.

⁴ Nyaope (also known as wonga or wunga) is a highly addictive, dangerous and destructive street drug unique to South Africa.

⁵ Wonga (also known as nyaope or wunga): see above.

Table 3.23: Tobacco use by students and staff at TVET colleges

Variables	Students		Staff		Test statistic
	n	%	n	%	
Cigarette use					
Yes	984	21.5	162	18	$\chi^2= 5.7$ $p = 0.016^*$
No	3 589	78.5	740	82	
Snuff use					
Yes	56	13.3	17	2.1	$\chi^2 = 60.7$ $p < 0.001$
No	366	86.7	780	97.9	
Chewing tobacco use					
Yes	28	7.2	6	0.8	$\chi^2 = 38.5$ $p < 0.001$
No	362	92.8	785	99.2	

* Pearson chi square

3.4.5. Exposure to incidents of crime and violence at TVET colleges

Table 3.24 summarises known incidents of violence at TVET colleges as well as other factors that impact on the security of institutions.

Overall, 5.3% of students and 5.6% of staff said a person had been shot at their institution; 5% of students and staff said someone had been killed at their institution; and 5.1% of students and 4.4% of staff said rape had taken place at their institution. In addition, 6% of students and 8% of staff were aware of sexual harassment occurring at their institution.

Assault, the carrying of weapons and gang activity appeared to be fairly widespread. The occurrence of assault at the institution was reported by 16.7% of students and 20.8% of staff, and a fight involving weapons by 11.8% of students and 8.7% of staff. Nearly one out of seven students (13.7%) and 8.7% of staff said they were aware of gangs operating at their institution.



Table 3.24: Incidents of violence at TVET colleges and related factors

Variables	Students		Staff		Test statistic
	n	%	n	%	
A person was shot at the institution					
Yes	281	5.3	51	5.6	$\chi^2 = 0.1$ p = 0.740
No	4 991	94.7	860	94.4	
A person was killed at the institution					
Yes	253	5	46	5	$\chi^2 = 2.3$ p = 0.886
No	4 978	95	862	95	
A person was assaulted at the institution					
Yes	870	16.7	189	20.8	$\chi^2 = 9.3$ p = 0.002
No	4 348	83.3	719	79.2	
A fight involving weapons took place on the premises					
Yes	614	11.8	79	8.7	$\chi^2 = 8.6$ p = 0.126
No	4 597	88.2	827	91.3	
A person was raped at the institution					
Yes	268	5.1	40	4.4	$\chi^2 = 0.8$ p = 0.373
No	5 217	94.9	862	95.6	
Students or staff found carrying weapons on the premises					
Yes	530	10.2	122	13.5	$\chi^2 = 9.7$ p = 0.137
No	4 675	89.8	785	86.5	
Someone was sexually harassed at the institution					
Yes	308	6	73	8	$\chi^2 = 6.3$ p = 0.012
No	4 911	94	834	92	
Awareness of gangs operating in the institution					
Yes	716	13.7	74	8.1	$\chi^2 = 22.5$ p < 0.001
No	4 504	86.3	835	91.9	

* Pearson chi square

3.4.6. Pregnancy and abortion among students and staff at TVET colleges

Data in Table 3.25 show that more pregnancies had occurred among staff and their partners than among students and their partners. This finding is consistent with the different age and marital profiles of students and staff.

Figure 3.4 indicates that more than two-thirds of staff (69.7%) had been pregnant or made a sexual partner pregnant, and this was the case for nearly one-third of students (31.7%).

Although most staff pregnancies (64.1%) were planned, one in three was unplanned. Among students only a quarter of pregnancies (24.6%) were planned. A higher proportion of students (or their partners) had undergone an abortion (16.1%) than was the case for staff and their partners (9.7%). Both these differences between staff and students were statistically significant ($p < 0.001$).

Table 3.25: Pregnancy and abortion among students and staff

Variables	Students		Staff			Test statistic
	n	%	n	%	n	
Have you ever been pregnant or made someone pregnant						
Yes	1 736	31.7	671	69.7	2 407	$\chi^2 = 504.8$ $p < 0.001$
No	3 439	62.8	265	27.5	3 704	
Not sure	301	5.5	27	2.8	328	
Was the pregnancy planned						
Yes	406	24.6	421	64.1	827	$\chi^2 = 317.6$ $p < 0.001$
No	1 242	75.4	236	35.9	1 478	
Have you or your partner ever had an abortion						
Yes	269	16.1	64	9.7	333	$\chi^2 = 15.6$ $p < 0.001$
No	1 406	83.9	596	90.3	2 002	

* Pearson chi square

Figure 3.4: Students who had been/made someone pregnant

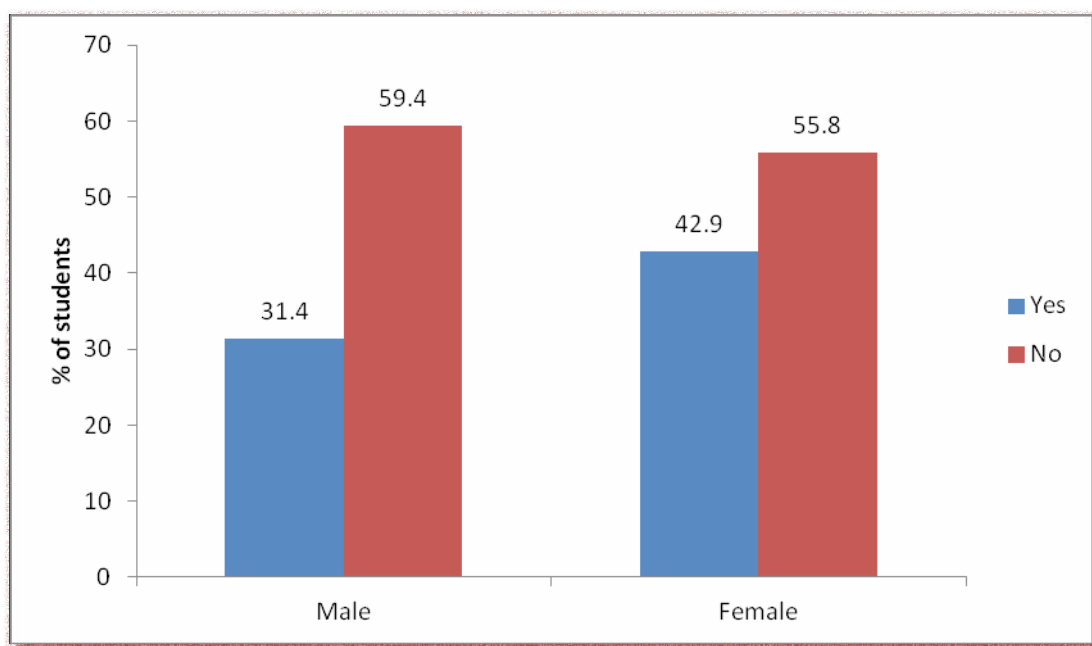
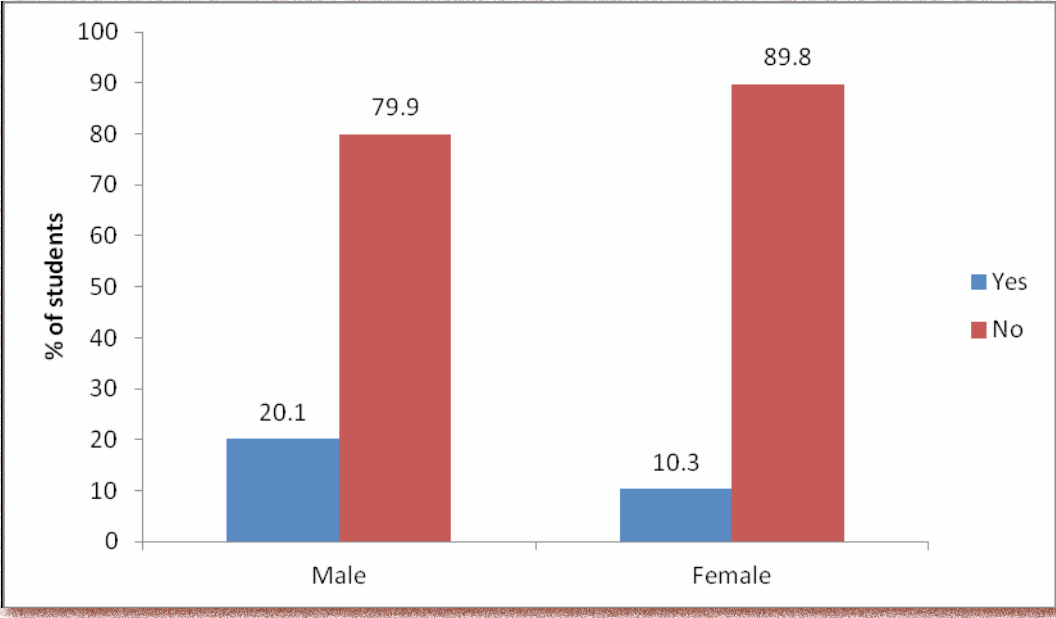


Figure 3.5: Students/partners who had undergone an abortion



3.5. Programmes and services for HIV prevention, treatment and care

3.5.1. Treatment services for students and staff at TVET colleges

Students and staff at TVET colleges used HIV services at government health facilities much more than other options.

More than three-quarters of students (77.3%) and staff (76%) indicated that they depended on a government health facility, while 10.6% of students and 35% of staff said they visited a medical doctor, and 7.6% of students and 32% of staff said they used a private health facility. A very small percentage of staff and students said that they used a traditional doctor or spiritual healer, and still fewer relied on family or friends.

Figure 3.6: Services utilised by students and staff for HIV healthcare

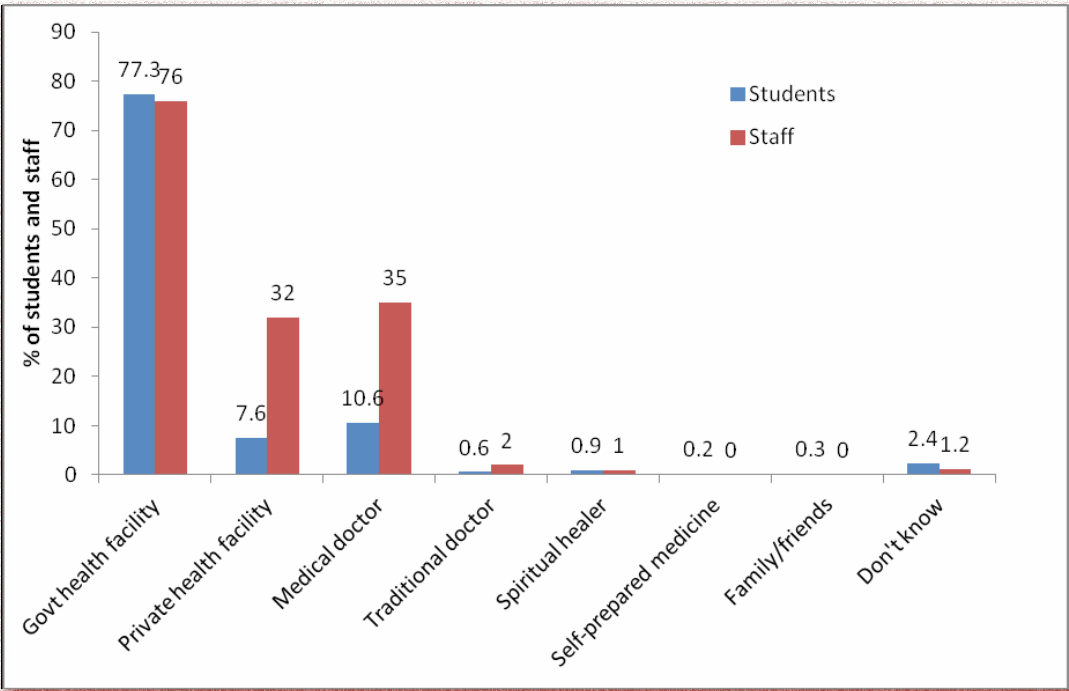
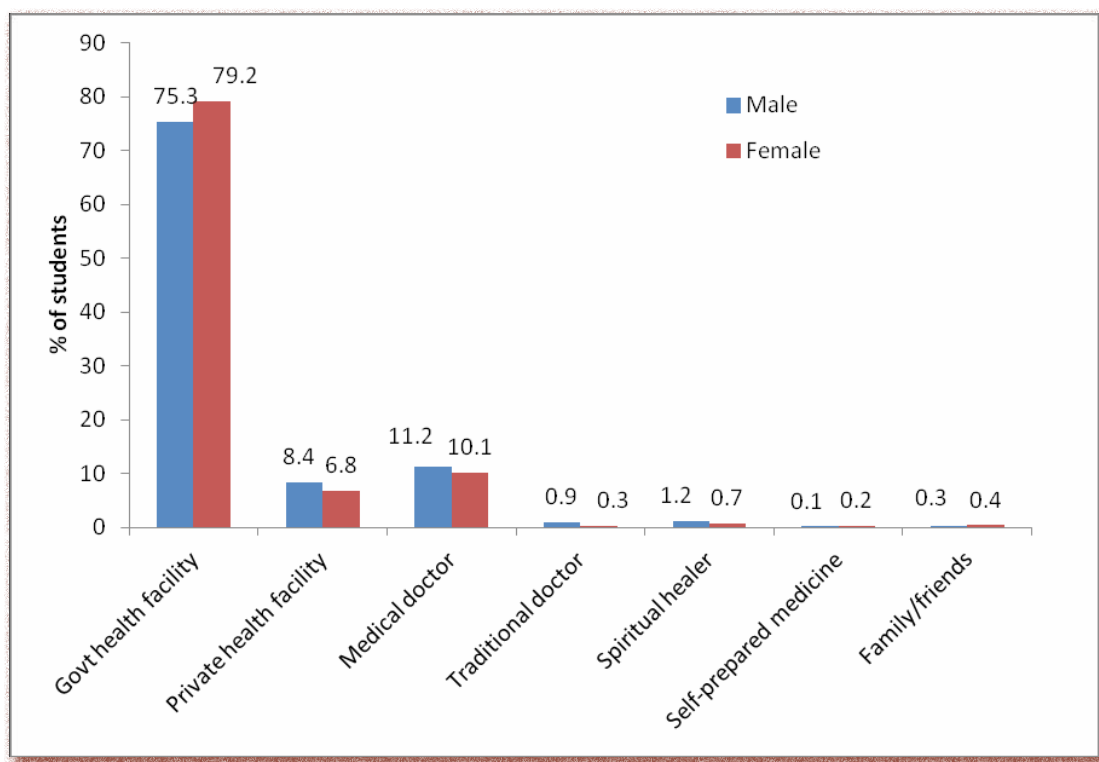


Figure 3.7 shows that there is very little difference between male and female students in the pattern of selection of health facilities for HIV services.

Figure 3.7: Services utilised by male and females students for HIV healthcare



3.5.2. Sources of HIV and AIDS information at TVET colleges

Staff members at TVET colleges were much more likely than students to utilise multiple sources of information on HIV and AIDS (Figure 4.8). A substantial proportion of staff members indicated that they sourced information from various forms of mass media (67.5% from television, 58.5% from radio, 55% from magazines and 48% from newspapers). The percentage of students utilising the mass media for HIV information was much lower, with 20.1% citing TV as a major source, 20% identifying radio, 18.6% mentioning magazines and 17% newspapers.

Health facilities were a major source of information for 18.1% of students and 47% of staff, while education institutions were mentioned by 15.3% of students and 33.3% of staff.

About one-third of staff (31%) said they used the Internet to access HIV and AIDS information and 14% resorted to the AIDS Helpline. The corresponding figures for students were 12% and 7.9%.

Social media have not taken off strongly at TVET colleges as major sources of HIV information: Facebook was used by 12% of staff and 8% of students, Twitter by 9.7% of staff and 6.4% of students, and Instagram by 5.8% of staff and 4.3% of students.

Figure 3.8: Main sources of HIV information for students and staff

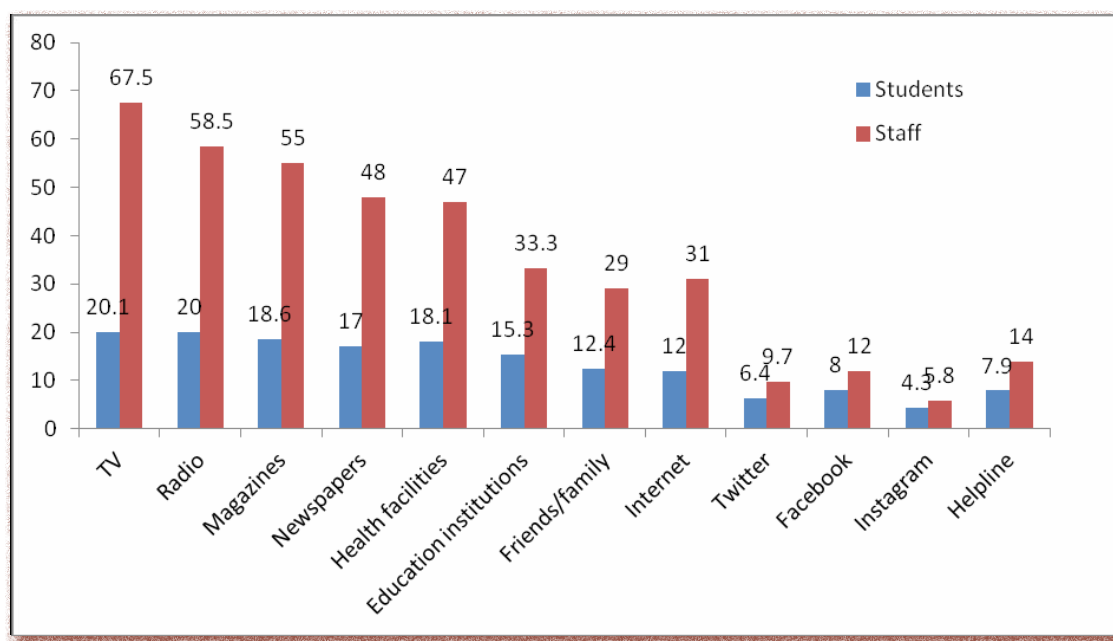


Table 3.26 summarises information on the HIV and AIDS programmes and services that TVET colleges provide for students and staff. A minority of respondents indicated that their colleges provided information on a range of subjects related to HIV prevention and treatment. A higher proportion of staff members than students identified HIV interventions and services undertaken by the institution. However, the most frequently mentioned interventions – provision of general information on HIV and AIDS, distribution of condoms, and HIV counselling and testing (HCT) – were only reported by 36.3%, 32.2% and 25.4% of staff members respectively.

Table 3.26: HIV programmes and services at TVET colleges

Variables	Students	Staff	n	%
	n	%		
Topic of information provided by colleges				
HIV and AIDS	1 254	22.2	364	36.3
Antiretroviral treatment (ARVs)	794	14	216	22
Prevention of mother-to-child transmission of HIV (PMTCT)	607	11	150	15
Sexually transmitted infections (STIs)	708	12.5	203	20.3
Tuberculosis (TB)	682	12.1	200	20
HIV counselling and testing (HCT)	689	12.2	232	23
Male medical circumcision (MMC)	465	8.2	129	13
Services provided by TVET colleges				
Condom distribution	1 134	20.1	323	32.2
ARV provision	397	7	54	5.3
HCT	876	15.5	255	25.4
TB testing	402	7.2	104	10.3
PMTCT	187	3.3	37	4
Testing for other STIs	336	6	78	8
Referrals	176	3.2	112	11

Table 3.27 shows that the extent of HIV education and information programmes provided at institutional level varied from one province to another. Provinces where a relatively high proportion of students and staff reported receiving information on a spectrum of HIV-related topics were Limpopo, Mpumalanga and the Western Cape. The proportion of respondents from these provinces who stated that they received information was consistently higher than average, no matter the specific topic.

The provinces where the lowest proportion of students and staff reported receiving information at their institutions on several HIV-related subjects were KwaZulu-Natal, the Northern Cape and Free State. Gauteng, the Eastern Cape and North West occupied the middle ground in terms of the percentage of respondents reporting that colleges provided information on HIV-related topics.

3.5.3. HIV programmes and services provided at TVET colleges

As indicated in Table 3.28, the availability of institution-based services related to HIV was not frequently reported by students and staff, regardless of province. However, two services – condom distribution and HCT – were more frequently said to be available at institutions.

In Limpopo, the Western Cape, Mpumalanga and the Eastern Cape, at least one out of five respondents said that the institution undertook the distribution of condoms. In Limpopo reports of on-campus condom distribution rose to 27.6%. However, the percentage of respondents who confirmed that condom distribution took place at their colleges dropped to 12.8% in the Free State and 12.9% in the Northern Cape.

The percentage of respondents reporting that HCT was offered by the institution was highest in Limpopo (26.2%), followed by Mpumalanga (20.6%) and the North West (14.4%). The lowest rates of confirmation of institutional HCT services were in KwaZulu-Natal (8.4%), the Northern Cape (8.9%) and the Eastern Cape (9.5%).

With very few exceptions, the percentage of respondents reporting that colleges made available ARV treatment services, TB testing facilities, PMTCT services and STI testing fell below – and usually far below – 10%.



Table 3.27: Content of TVET information programmes by province

Information provided by institution about:	EC	FS	GP	KZN	LP	MP	NC	NW	WC										
	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n
HIV and AIDS	93	22.1	114	15.8	253	17	207	16.3	232	32.3	142	27.4	53	15.1	85	19.1	168	23.1	
Antiretrovirals (ARV)	52	12.4	66	9.2	148	10	116	9.1	135	18.8	92	17.7	28	8	52	11.7	105	14.4	
PMTCT	50	11.9	51	7.1	106	7.1	76	6.0	93	13	73	14.1	23	6.6	47	10.6	88	12.1	
STIs	49	11.7	63	8.8	137	9.2	86	6.8	110	15.3	93	17.9	29	8.3	50	11.2	91	12.5	
TB	59	14	61	8.5	118	8	87	6.8	111	15.5	83	16	22	6.3	49	11	92	12.7	
MMC	41	9.8	58	8.1	132	8.9	80	6.3	127	17.7	84	16.2	29	8.3	52	11.7	86	11.8	
Other	27	6.4	43	6.0	89	6	61	4.8	63	8.8	54	10.4	14	4	44	9.9	70	9.6	

Table 3.28: HIV-related services provided to students and staff at TVET colleges by province

Services provided by institution	EC	FS	GP	KZN	LP	MP	NC	NW	WC										
	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n	%	n	%	%
Condom distribution	84	20	92	12.8	203	13.7	172	13.5	198	27.6	109	21	45	12.9	70	15.7	161		22.1
ARV provision	40	9.5	28	3.9	74	5	61	4.8	74	10.3	30	5.8	19	5.4	29	6.5	42		5.8
HCT services	40	9.5	88	12.2	167	11.3	107	8.4	188	26.2	107	20.6	31	8.9	64	14.4	84		11.6
TB testing	29	6.9	50	6.9	62	4.2	54	4.2	65	9.1	29	5.6	15	4.3	32	7.2	66		9.1
PMTCT	19	4.5	20	2.8	38	2.6	27	2.1	22	3.1	8	1.5	11	3.1	14	3.1	28		3.9
Testing for STIs	24	5.7	26	3.6	69	4.6	48	3.8	60	8.4	36	6.9	12	3.4	25	5.6	36		5
Referrals	37	8.8	7	1	27	1.8	22	1.7	23	3.2	20	3.9	4	1.1	15	3.4	21		9.6

This is the first national survey on knowledge, attitudes and behaviours related to HIV and AIDS among students and staff at TVET colleges in South Africa. It complements an earlier study of a similar nature conducted at universities in South Africa. The current study concludes with recommendations for HIV prevention interventions to be implemented at TVET colleges. It also provides baseline data against which to measure the impact of future interventions that aim to increase knowledge and modify attitudes and behaviours.

In some cases the results have confirmed what is already known about HIV and AIDS at higher education institutions (HEIs) while in other cases they have pointed to additional challenges faced by TVET colleges.

The implementation of HIV prevention interventions in the wider society benefits the TVET sector and could have affected or contaminated the findings of this study. For example, at the time of the study, South Africa had established the world's largest ARV treatment programme, and distribution of condoms and provision of HCT in the public health sector were progressing well. Students and staff at TVET colleges would have had access to these interventions through public health facilities in the communities where they live, study or work.

In addition, HEAIDS began to introduce HIV prevention interventions at TVET colleges during the course of the survey. For example, the First Things First HCT campaign was launched in the TVET sector in Cape Town around the time of this study.

The survey findings presented in Chapter 3 are discussed below.

4.1. Knowledge of HIV and AIDS

4.1.1. Awareness of AIDS and TB

This survey established that awareness of HIV and AIDS among students and staff at TVET colleges was high (87% of all respondents). Awareness levels are important as they provide the foundation for implementing effective interventions to build knowledge, attitudes and behaviours that reduce the risk of infection.

In terms of TB, 29% of the respondents said they had undergone a TB test and 21% of this group reported that their results were positive. TB testing is only undertaken if an individual displays symptoms of TB, has been in close contact with someone with active TB or is HIV-positive. Given the fact that South Africa has the third highest number of active TB cases in the world (SANAC, 2012), it was surprising that 73% of respondents said they had not been exposed to anyone with TB. This could indicate a low level of awareness and risk perception in relation to TB. It is, however, encouraging that 87% of students and staff at TVET colleges knew that TB can be cured.

4.1.2. Knowledge about HIV and STI transmission

This survey suggests that the country has progressed well in communicating messages on HIV prevention and that the TVET sector has benefited from information, education and communication campaigns in the wider community. It is critical to state that the results show that this information is not necessarily being accessed or provided as part of college intervention programmes.

Knowledge about HIV transmission among students and staff was exceptionally high: approximately 96% of students and staff knew that HIV could spread to men and women through unprotected sex. A high proportion of respondents also knew that the presence of STIs increased a person's risk of acquiring HIV and that HIV-positive people who did not look sick could transmit the virus. A large majority of both staff members and students knew that

AIDS cannot be cured. However, it is a matter of some concern that 16.2% of respondents still believed that AIDS could be cured.

The findings of this study in respect of knowledge about HIV transmission were similar to those of the 2012 HSRC study in which: 61% of participants agreed that having fewer sexual partners reduced the risk of acquiring HIV; nearly 80% stated that always using a condom prevented HIV acquisition; and 84% understood that a healthy-looking person could have HIV. In the HSRC 2012 study, 21% of participants believed that AIDS could be cured and another 10% were uncertain about this, meaning that only 69% had the correct information that HIV cannot be cured (Shisana, Rehle et al, 2014).

4.1.3. Knowledge of where to get condoms

The correct and consistent use of male condoms remains central to programmes designed to prevent HIV and other STIs. This practice reduces the risk of HIV transmission by almost 100%. Consistent and correct condom use is also an effective form of contraception. Condom use by young people in South Africa has increased significantly over the past two decades (Reddy et al, 2013).

In this study more than 90% of students and staff knew that people can reduce their risk of HIV infection by using a condom each time they have sexual intercourse. These findings are similar to those contained in other national surveys. The third National HIV Communication Survey (NCS), in particular, found very high levels of knowledge about condom use (Johnson et al, 2013).

In the TVET survey, knowledge of where to get condoms was extremely high among both students and staff. However, this should not be construed as indicating that condoms were conveniently available on most campuses as further questions indicated this was not the case.

4.1.4. Knowledge of preventive use of antiretroviral medicines

South Africa currently has the largest ARV treatment programme in the world. This programme has not only extended the lives of those on treatment and contributed to an increase in life expectancy, but it also plays a vital role in preventing the spread of HIV. The ARV programme includes pregnant women and reduces infections in babies and young children (Shisana et al, 2014).

Most respondents in the current study (80% of students and 88.7% of staff) had heard that ARVs can be taken by HIV-positive pregnant women to reduce the risk of infecting the baby. Only 60% of students and 75% of staff had heard post-exposure prophylaxis for women who had been raped.

4.2. Attitudes to PLHIV and LGBTI

4.2.1. Attitudes to people living with HIV

Stigma and discrimination against people living with HIV (PLHIV) remain major barriers to effective HIV prevention and the provision of treatment, care and support services (Shisana, Rehle et al, 2014). This section of the report addresses aspects of general stigma in relation to PLHIV.

Although a majority of students and staff at TVET colleges had positive attitudes to PLHIV a significant minority of students and staff did not want to be associated with HIV-positive individuals or share a house with them. These negative attitudes were more likely to be held by students than staff, with male students and white students holding more negative views than their female and African counterparts.

More students (20%) than staff (14%) expressed the opinion that people who become infected with HIV are promiscuous. Overall, about one out of five students felt that HIV-positive people

should not be allowed to socialise with HIV-negative people, and in some geographic areas this view was even more common.

It was somewhat surprising, in the light of the above views, that 89% staff and 88% of students said if a family member had HIV their relationship with that individual would be good. Students' expectations of sympathy should they test positive were much lower: only 38% thought they would be supported by friends, suggesting that institutions have a way to go in order to achieve a climate of openness about HIV status.

In the HSRC 2012 study, the majority of respondents also expressed positive attitudes in response to five of the six stigma questions that were investigated. The range of positive responses was 79% to 92%) with the question "Would you be willing to care for a family member with AIDS?" receiving the highest proportion of affirmative responses. However, respondents were ambivalent to the question, "Would you want to keep the HIV-positive status of a family member a secret?" Exactly half of them said they would favour secrecy. Only one-sixth of respondents agreed that it would be "a waste of money to train or give a promotion to someone with HIV/AIDS" (Shisana, Rehle et al, 2014).

4.2.2. Attitudes to treatment for HIV and AIDS

Students and staff at TVET colleges generally had positive attitudes to treatment for HIV and AIDS. Almost the same proportion of students and staff (79% and 77% respectively) said they trusted that HIV treatment worked, while 80% of students and 82% of staff agreed that ARVs would keep PLHIV alive, and 83% rejected the idea that ARVs were poisonous.

However, the above figures also indicate that one in five students and staff still had no belief in and understanding of a proven method of treatment for a disease that is otherwise fatal. More than a quarter of all students and staff (28%) incorrectly believed that you can share HIV treatment with your partner. This is a worrying finding considering that students and staff at TVETs are more literate than the general population.

4.2.3. Attitudes to condom use

The survey revealed that a high proportion of students and staff held negative attitudes to condom use. Four out of ten respondents (41% of students and 43% of staff) said condoms felt unnatural with three out of 10 (31% of staff and 30% of students) said condoms changed the experience of orgasm. About one in five respondents expressed the view that a woman loses a man's respect if she asks him to use a condom.

These results paint a troubling picture as the condom is still the most effective and cost-effective way to prevent sexual transmission of HIV. These negative attitudes to condom use may account for the fact that consistent condom use with all sexual partners is still not universally observed by students and staff at TVET colleges.

They may also be related to a broader decline in condom use in recent years, as reported in HSRC studies in South Africa and by other researchers in African countries such as Cote d'Ivoire, Niger, Senegal and Uganda (UNAIDS, 2013c).

Another possible factor behind the recent decline of condom use in South Africa is the reduction in prevention efforts focusing specifically on condom use. In addition, some researchers suggest that reporting on condom use was overestimated in the past and is now more accurate as people are more open to discussing the matter (Johnson et al, 2012). Another possible explanation is that the general availability of ARV treatment for AIDS might have led to "behavioural disinhibition" or increased sexual risk-taking because the consequences of acquiring HIV are no longer as drastic (Shisana, Rehle, Simbayi et al, 2014).

Campbell (1997) summarises findings for sub-Saharan Africa that relate negative attitudes to condom use to cultural factors, such as the desire for children and female sexual compliance, and their links to improved economic status. Lula and Greer (1991) found that only a minority of Ugandan students saw the condom as an effective method of protection against HIV: most viewed it as unsafe or an encouragement to promiscuity. Abdool-Karim et al (1992:109), in a qualitative study exploring high school students' reasons for not using condoms, found students felt they did not understand condom use well enough and condoms were not available when required. Distribution of free condoms has increased dramatically since this study was undertaken and IEC activities on condom use also peaked subsequent to this study, so the factors cited may no longer be as important.

4.2.4. Attitudes to prevention of STIs

The high incidence of STIs constitutes a major public health problem in South Africa. Studies have reported that having an STI multiplies the risk of acquiring HIV by a factor of between two and eight.

Participants' attitudes to requesting their partner to undergo STI testing revealed a high level of uncertainty or anxiety about the impact this request might have on their relationships with their partners. About four out of 10 participants did not know what the partner's response would be if they were to request him or her to undergo an STI test. More than a third thought it was likely or highly likely that the partner would suspect that the participant had cheated on him/her or was accusing him or her of giving the participant an STI. Partner notification, testing and treatment is a critical element in reducing the incidence of STIs and these responses indicate why it is difficult to achieve high rates of partner testing.

4.2.5. Attitudes to LGBTI people

There have been multiple media reports in recent years about violence aimed at LGBTI individuals in South African communities. These incidents suggest that discrimination against LGBTI people is real and potentially dangerous, making it important to understand attitudes to LGBTI people at TVET colleges.

The survey found generally positive attitudes to LGBTIs among both students and staff. A sizable proportion of respondents among both students (8%) and staff (5%) identified themselves as homosexual.

However, only 53% of students and 61% of staff said they considered their institution to be a safe environment for LGBTIs. This low level of perceived safety for LGBTIs on TVET campuses may reflect events in South African communities where there have been several reports of extremely violent treatment of LGBTIs, including "corrective" rape.

A study on students' attitudes to gay and lesbian issues at a Michigan college in the US found five key patterns. Firstly, the majority of students became more accepting of lesbians, gay men and bisexual people during their four years at college. About 60% to 70% of students who had entered college with negative attitudes became more accepting and 50% of those entering with ambivalent attitudes displayed more positive attitudes at the end of four years. Secondly, on average, women entered college with more accepting attitudes than men and also increased their acceptance over four years more than men. Thirdly, getting to know lesbians, gay men, and bisexual people was a primary mechanism through which students' attitudes changed. Fourth, engagement with this topic through curricular and co-curricular programming helped to establish norms of respect and promoted thoughtful consideration of these issues. Finally, participation in student religious groups was associated with an absence of change in students' attitudes on this topic (Kardia, 1995).

4.3. Behaviours that put students and staff at risk of HIV and STIs

4.3.1. Sexual partnerships

More than a third of respondents in this study reported that they had a sexual partner who was at least two years older than them, and almost 15% of this group indicated they had a partner who was at least five years older than them. Only one in five respondents reported dating people who were the same age as themselves.

There was a significant difference in age-disparate relationships between males and females among both staff and students: while young women in such relationships mostly had partners older than themselves, young men tended to have partners younger than them.

Age-disparate relationships have been widely regarded as a factor that increases the risk of acquiring HIV, especially among girls (Shisana, Rehle, Simbayi et al, 2014). Young people who have partners five or more years older than themselves are more exposed to HIV because HIV prevalence generally increases with age (Shisana, Rehle, Simbayi et al, 2005). In addition, young people with older partners are at increased risk of engaging in risky sexual activity, contracting an STI, not using contraception and becoming pregnant (Kirby, 2007).

Age-disparate relationships have been found to be driven strongly by poverty, as they may enable young girls to obtain from an older partner the material things they aspire to (Leclerc-Madlala, 2008; loveLife, 2012). The promise of gifts and other financial and social benefits, including the potential for upward social mobility, can impact on girls' age of sexual debut and motivate girls to engage in age-disparate sexual relationships, transactional sex or marriage at a young age (UNICEF, 2013). Where HIV prevalence is high, these factors combine with increased biological vulnerability, limited knowledge about HIV and low risk-perception to influence the choices girls make and expose them to a high risk of infection.

4.3.2. Condom use

In this study, students and staff alike reported low rates of condom use at last sex where this involved a regular partner (55.1% for students and 34.5% for staff) or a sex worker (51% for students and 33% for staff). However, condom use at last sex was reported to be high with a non-regular partner (75.5% for students and 54.8%).

Levels of condom use at last sex with a (regular) partner were similar to those reported in the HSRC national household survey conducted in 2012. The latter showed a decline in condom use at last sex since the previous household survey conducted in 2008 (Shisana, Rehle et al, 2012) and the TVET study confirmed the reduced rate of use. Condom use at last sex among TVET students was also comparable to that measured among students at HEIs in the HEAIDS 2010 study, which found that 62% of male students and 58% of female students had used a condom the last time they had sex.

Regardless of the type of sex partner, a low proportion of students and staff reported using condoms "every time" they had sex. However, the figure was still higher than that reported in the HSRC 2012 household survey (Shisana, Rehle et al, 2012). In contrast to the household survey, a high proportion of TVET students and staff also said that they used condoms "almost every time". Respondents' interpretation of the phrases "almost every time" and "sometimes" in relation to condom use is subjective and this may account for some of the variation between surveys.

4.3.3. Substance use

Alcohol use among students at TVET colleges seems to be relatively high, particularly among male students, and higher than the rates reported in the YRBS where about a 50% of high school learners reported ever using alcohol (Reddy et al, 2013).

In the HEAIDS 2010 study of HEIs, alcohol use was reported by 35% of students, 14% of academic staff, 21% of administrative staff and 24% of service staff. This study also established a pattern of binge drinking among some students and staff.

Specific patterns of alcohol use were not explored in the TVET study. However, it must be noted that excessive alcohol use is widely recognised as a risk factor for sexual risk-taking which may potentially lead to HIV/STI transmission. It is also linked to other risky behaviours such as drunk driving and violence.

About one in five students reported lifetime use of cigarettes with a slightly lower rate reported by staff members. Smokeless tobacco use, involving snuff and chewing tobacco, was more than five times as high among students as it was among staff members. Similar rates of tobacco use to students at TVET colleges were found in the YRBS among high school learners. In the latter study, the rate of lifetime use of cigarettes was 27%, with coloured learners reporting significantly higher rates (54%). Use of smokeless tobacco products was reported by 10.6% of participants (Reddy et al, 2013).

Additional substance use among students and staff at TVET colleges was very low except for marijuana or dagga. One in 10 TVET students and 5% of staff members reported ever using dagga. Data from the YRBS showed about 12% of high school learners had used dagga at some point and 9% in the past month.

Only about 1% of students and 0.6% of staff reported that they had used hard drugs such as mandrax⁶, tik⁷, 1% nyaope⁸, cocaine or crack and ecstasy among others. The survey also found extremely low use inhalants such as glue among TVET students and staff members. In contrast, in the YRBS the reported use of hard drugs was about 5% (Reddy et al, 2013).

4.3.4. Male circumcision

In this study the self-reported level of male circumcision was high, with more than half of all male respondents reporting that they had undergone circumcision. There was a clear preference for circumcision to be performed in hospital settings by medical doctors. There was some variation in support for medical circumcision across provinces but the only province in which traditional circumcision was preferred to medical circumcision was the Eastern Cape. The HSRC 2012 survey, with its more varied sample, painted a less clear-cut picture in terms of circumcision preferences.

Men are at highest risk of acquiring HIV in their twenties and thirties. It is therefore encouraging to see support for MMC among students, most of whom are just entering this high-risk age group and have a critical window for risk-reduction. High-level advocacy, such as the launch of a national MMC campaign by former Deputy President Kgalema Motlanthe, in his capacity as chair of the South African National AIDS Council (SANAC), may have helped generate favourable attitudes to MMC. TVET colleges could play an important role in converting the sympathetic attitudes of students into action on MMC, thereby contributing to the achievement of the target in the National Strategic Plan for HIV, TB and STIs 2012-2016 (SANAC, 2011).

4.3.5. Crime and safety at TVET colleges

TVET colleges appear to be vulnerable to the violent crime that characterises South African society. About 5% of respondents reported the occurrence of serious criminal incidents on campus, including shootings, killings and rape. These incidents highlight the need for TVET

⁶ Mandrax was initially marketed as a sedative or sleeping tablet by French pharmaceutical giant Roussel Laboratories. It turned out to be highly addictive and was banned in 1977.

⁷ Tik is the slang name given to the stimulant drug, Methamphetamine. Tik and all other forms of crystal meth are stimulants - they increase activity between the brain and the central nervous system.

⁸ Nyaope (also known as whoonga or wunga) is a highly addictive, dangerous and destructive street drug, unique to South Africa.

colleges to adopt a stance of zero tolerance of behaviours that help create an environment in which violence is an ever-present possibility.

This study revealed a high level of interpersonal violence on TVET college campuses. About 17% of students and 21% of staff said a person had been assaulted at their institution. Fights involving the use of weapons and the carrying of weapons by both staff and students on campuses were mentioned by about one in 10 respondents. The reported existence of gang activity on campuses may be a factor and there is a possibility that gang turf wars may be spilling into campuses.

The findings of the present study are consistent with other national studies. The YRBS reported that about 13% of high school learners said they carried a weapon to school, 7% brought a gun to school and about 27% admitting being involved in a physical fight while at school.

In the HEAIDS higher education sector study (2010), although the majority of students said they felt safe from physical harm at their institution, 60% of service staff and 71% of academic staff did not feel safe. About half of the staff and 18% of students also agreed with the statement “violent crime, where people are physically injured, is a serious problem at this institution”.

4.3.6. Pregnancies and abortions

Predictably there was a higher rate of pregnancy among staff members and their partners (nearly 70%) than there was among the younger and largely unmarried population of students (nearly 32%). The implications of the latter figure must be appreciated: nearly one-third of students have dealt with pregnancy by the time they enter a TVET college and most of these students have become parents.

The rate of unplanned pregnancies among staff members was high (about 36% of all those who had been pregnant or made someone pregnant) while it was extremely high among students (75%). This provides further evidence of the lack of safe sex practices among both students and staff and a failure to utilise alternative contraceptive methods.

About 16% of students who had been pregnant (or their pregnant partners, in the case of male students) elected to have an abortion, compared to about 10% of staff members.

4.4. Programmes and services for HIV and AIDS at TVET colleges

4.4.1. Services for HIV and AIDS treatment

TVET campuses currently do not have capacity to provide HIV-related healthcare or students and staff.

About three-quarters of students and staff said that they relied on government health facilities for HIV-related healthcare. About three out of 10 staff members also indicated that they used private health facilities and the same proportion said they consulted private medical doctors. It appears, therefore, that some staff members may use a mixture of public and private healthcare.

Only a very small percentage of students and staff acknowledged utilising traditional and spiritual healers for HIV-related care.

The survey highlighted major gaps in the provision of services on TVET campuses. Most campuses did not even undertake the most basic intervention, ensuring that supplies of condoms were available to students and staff. Only one in five campuses provided condoms to students and only about a third of staff reported having access to condoms on campus.

HCT provision on campuses was extremely low. Only a quarter of staff members and about 16% of students reported that this service was available on campus. This is clearly an area that will require special attention. Findings in the NCS 2012 suggest that condom distribution and HCT coverage is much higher in the general population than for TVET students and staff on campus.

4.4.2. Sources of HIV and AIDS information

Communicating information about HIV, TB and STIs is a fundamental aspect of prevention programmes. If people are aware of the severity of HIV and TB and the steps they can take to protect themselves, they will be more inclined to take risk-reduction measures. It is therefore essential to assess which sources of information are useful for encouraging the TVET population to take HIV seriously (Shisana, Rehle et al, 2014).

The findings from this study show the most important source of information for both students and staff was television followed by radio, magazines, health facilities, and educational institutions.

The very limited role of TVET colleges in making information available to students is a striking fact highlighted by this study.

Staff members were more than twice as likely as students to use mass media as a source of health information and the proportion of students relying on mass media information – about one in five – was surprisingly low.

This level of mass media use by students is in sharp contrast to the findings of the NCS 2012 which reported that 82% of the population aged 16-55 years had been exposed to one or more HIV communication programme, mostly through the mass media, and that exposure to these programmes had resulted in improved knowledge of HIV as well developing attitudes conducive to HIV prevention and changing behaviour patterns (Johnson et al, 2013).

The current study's findings on utilisation of mass media information also differed substantially from the HSRC 2012 results where 67% pre-adolescents, 51% of youth and 52% of young adults indicated that information on television had encouraged them to consider HIV as a serious condition. Radio programmes came a distant second to TV and print media third.

The HSRC study suggested that leaflets and posters did not play as important a role as mass media in changing perceptions about the seriousness of HIV. On the other hand, talking to health workers and family members emerged as an important factor in changing perceptions among youth and adults (Shisana, Rehle et al, 2014).

An examination of the data across the three HSRC surveys shows that there was a decline from 2005 to 2012 in the effectiveness of electronic media in getting people to take HIV seriously (Shisana, Rehle et al, 2014). The 2012 NCS also noted a drop in the reach of mass communication programmes on HIV between 2009 and 2012 (Johnson et al, 2013).

Social media have not taken off strongly as sources of HIV-related information at TVET colleges. The main source of online information was the Internet which was used by 31% of staff and 12% of students. Social media utilisation was as follows: Facebook - 12% of staff and 8% of students; Twitter - 10% of staff and 6% of students; and Integral - 6% of staff and 4% of students.

The telephonic AIDS Helpline was used by 14% of staff and 8% of students. Staff reported much higher utilisation of HIV prevention information from healthcare facilities than students (47% versus 18%). This is unexpected since primary healthcare facilities in South Africa provide free services and should be equally accessible to students and staff.

4.5. Strengths and limitations of the study

4.5.1. Strengths

This was a large study, covering all 50 TVET colleges and drawing a sample that ensured representation of the entire TVET population. It surveyed 70% of the college campuses in nine provinces. The intended sample was 5 400 but data were collected from 6 654 respondents, yielding a response rate of 123%. The sample comprised 5 651 (85%) students and 1 003 (15%) staff.

The multi-stage cluster sampling methodology which was used to select a sample of campuses and classes within campuses ensured representativeness of the target sample. Both the sample size and its representativeness made it possible to generalise the results to the entire first-year student population at TVET colleges.

As the first such study in the TVET sector, this study benefited from the design of 2008-2009 higher education sector HIV study. It utilised and validated some of the tools employed in the earlier survey.

4.5.2. Limitations

The results of this survey should be read with caution in at least three areas. An important limitation, common to nearly all KAB surveys related to HIV and AIDS, is that they are based on self-reporting by respondents. Self-reporting may be biased by participants' ability to recall details and a tendency to provide socially desirable answers to questions that relate to private matters, such as sexual or addictive behaviours. Questions used in this study to elicit self-reporting on intimate or socially stigmatised behaviours had been validated in other scientific surveys dealing with similar issues (Shisana, Rehle, Simbayi et al, 2009).

The number of staff members and students who described themselves as Indian or of Indian origin was small (68 in total) and comprised only 1% all respondents. Since an estimated 2.1% of TVET students are Indian, this group was under-represented in the study sample. As a result it was not possible to make statistically significant comparisons of data for Indian participants with data obtained for other groups in the study. It was also not possible to generalise the findings as they pertain to Indian students.

The reluctance of the Indian or Asian group to participate in surveys has been observed in previous studies in South Africa, including all the three previous HSRC surveys. It appears that members of this group believe that they are not vulnerable to HIV and that HIV is not a problem in their community (Shisana, Rehle, Simbayi et al, 2009).

The third major limitation arises from the fact that the survey was cross-sectional in design and this does not allow us to establish cause and effect. Even when two factors seem to be associated (for example, participation in HEAIDS programmes and lower HIV rates) cross-sectional studies cannot indicate whether this relationship is a causal one. This is partly because these studies do not show the sequence of events and also because they cannot establish whether a third variable (for example, high risk perception due to direct contact with PLHIV) is driving both factors that seem to be linked (that is, both the level of participation in HEAIDS programmes and lower HIV rates).

Determining the impact of a diverse national prevention campaign such as HEAIDS is difficult in itself. Doing so in an environment where several similar HIV interventions operate makes it even more difficult. Changes in HIV knowledge, attitudes and practices over time are due to the combined effect of various prevention initiatives and it is extremely difficult to ascribe a particular result to a specific intervention.



5.1. Conclusions

The survey has provided the most comprehensive information available to date on knowledge, attitudes and behaviours of students and staff at TVET colleges that might put them at risk of acquiring or transmitting HIV.

The survey corroborates earlier studies showing that young South Africans are at considerable risk of HIV infection and underscores the importance of initiating and/or strengthening relevant programmes at TVET colleges to mitigate the impact of HIV and AIDS on students and staff. This survey provides a baseline against which the progress of the above programmes can be measured.

The positive findings from this study were:

- Awareness of AIDS among students and staff at TVET colleges was high: 87% of respondents said they had heard of AIDS.
- Knowledge about HIV transmission among students and staff was exceptionally high as was knowledge of where to obtain condoms.
- Awareness and knowledge about medical treatment of AIDS were high.
- Most students and staff at TVET colleges held positive attitudes to AIDS treatment and had confidence in ARV medication.
- The majority of students and staff at TVET colleges had positive attitudes to PLHIV.
- Most respondents also showed acceptance of LGBTI individuals.
- Condom use at last sex was with a non-regular partner was high.
- The self-reported level of male circumcision was high and there was widespread support for hospital circumcision done by doctors.
- Substance use among students and staff at TVET colleges was very low except for alcohol and marijuana or dagga.

The negative findings from this study were:

- A substantial minority of students and staff at TVET colleges still had negative attitudes to associating with HIV-positive people.
- Negative attitudes to condom use are common among students and staff at TVET colleges.
- Condom use with a regular sex partner was low and condom use overall was not consistent enough to provide effective protection against HIV, STIs and unwanted pregnancy.
- There were high levels of unplanned pregnancy among TVET college students and their partners, suggesting a lack of knowledge about contraception, high levels of unprotected sex and lack of effective access to contraceptive products.
- There were relatively high levels of alcohol use among both students and staff.
- Both students and staff depend overwhelmingly on government health facilities for AIDS services and very few have on-campus options.
- Students at TVET colleges had very limited access to useful information about HIV and related matters. On-campus information resources were underdeveloped and utilisation of mass media and online media was low.
- Social media have not taken off strongly as a major source of HIV information at TVET colleges.

5.2. Recommendations

There have been no substantial, structured interventions in relation to HIV and AIDS, TB and STIs for students and staff in the TVET sector and this is a serious missed opportunity in the national effort to overcome the joint epidemics that have devastated this country.

There is no time to be lost by HEAIDS in making up for the omissions of the past and the findings of this study provide guidance for a major scale-up of biomedical, behavioural, social and structural interventions designed specifically to impact on the TVET community.

Separate prevention strategies and interventions must be developed for student and staff populations. A framework, similar to the HIV Monitoring and Evaluation Framework for Higher Education, 2010, should be developed for TVET colleges.

The interventions that are recommended below are linked to the four strategic objectives of the NSP 2012-2016:

- Addressing social and structural factors across various sectors that drive these epidemics, influence their impact, and affect the way we care for affected people.
- Preventing new HIV and TB infections, as well as other STIs, through a combination of biomedical, behavioural, social and structural interventions.
- Sustaining health and wellness, primarily by access to quality treatment, care and support services, and developing programmes that focus on wellness.
- Protecting the human rights of people living with HIV, primarily through ending stigma, discrimination, human rights violations and gender inequality (NSP, 2012).

5.2.1. Biomedical HIV prevention and treatment interventions

The TVET sector, supported by HEAIDS, should scale-up and sustain successful biomedical interventions for prevention and treatment of HIV, including:

- *Providing free ARV treatment:* Wellness clinics at TVET colleges should be upgraded to support an HIV prevention and AIDS treatment strategy aimed at reducing the rate of infection in the country by 50% and providing free ARV treatment to 80% of those who need it. The TVET sector should be a major extension of the national AIDS treatment programme.
- *Testing and treating for HIV and TB:* All wellness clinics at TVET colleges should be able to test for HIV and TB and treat students and staff for HIV-related conditions as well as TB. This measure is necessary because of the extraordinarily high rate of TB in South Africa and the fact that 70% of TB patients in this country are co-infected with HIV.
- *Developing contraceptive services:* Priority should be given to developing contraceptive services at college wellness clinics. These services should offer advice and counselling to students as well as a full range of contraceptive options and should promote dual protection.
- *PMTCT programme:* Wellness clinics at TVET colleges should also have the capacity for services to reduce the rate of mother-to-child transmission of HIV through appropriate support to pregnant students and staff. This service goes hand-in-hand with the national campaign to encourage all pregnant women to initiate antenatal clinic visits as soon as they suspect that they are pregnant in order to receive optimal care.
- *Medical male circumcision:* The TVET sector should develop a specific MMC programme or strengthen its ties with the *Brothers for Life* programme to encourage men at TVET colleges to undergo MMC.
- *Post-exposure prophylaxis following rape:* HIV prevention strategies at TVET colleges must promote awareness of PEP for rape survivors, provide the service on campus and facilitate access to local back-up services for after-hours and emergency use.

5.2.2. **Behavioural and social HIV prevention programmes**

Colleges in the TVET sector, with the support of HEAIDS, should develop, strengthen and implement social and behavioural change communication (SBCC) programmes for students and staff members. The aim of these would be to establish or sustain high levels of knowledge about HIV, TB and STIs, build positive attitudes to proven ways to prevent infection, and encourage risk-reduction behaviours.

SBCC programmes should be designed so that they address social and structural factors in the TVET sector that drive the HIV, STI and TB epidemics and affect the way we care for affected people. Communication is a dynamic process and it can change the social environment in which individuals make personal choices. It can also be used as a tool of advocacy intended to secure changes at policy level that enable individuals to adopt healthier practices.

TVET SBCC programmes should include, but not be limited to, the following:

- *First things First campaign*: Introduce or sustain the *First Things First* HCT campaign, which includes screening for TB and is aimed at students and staff across the higher education sector. While the direct preventive impact of HCT has not been firmly established, it is a critical entry point to services for HIV- positive and HIV-negative people.
- Use of social media for HIV education: Explore innovative ways to use social media and communication and technology in ways that appeal to students and staff at TVET colleges and support health-seeking behaviour.
- Rights-based approaches: Develop interventions that will help students and staff at TVET colleges to overcome their prejudices towards PLHIV (where applicable) and build universal respect for the rights of PLHIV.
- Increased condom use: Introduce relevant interventions to change attitudes to condom use, especially during sex with a regular partner and with sex workers. An evidence-based, diversified and comprehensive response to the HIV epidemic must include a combination of prevention interventions designed to encourage both fidelity and condom use (Shisana, Rehle, Simbayi et al, 2014).
- Coloured condoms: Introduce coloured and scented condoms at TVET colleges to revive interest in condom use and find ways to address the personal and social attitudes that impact negatively on condom use.
- Prevention of STIs: Introduce interventions to increase knowledge of and change attitudes to preventing STIs. Address social behaviours that put students and staff at risk of HIV, STI infection and reinforce norms and behaviours that are protective.
- Age-disparate relationships and intergenerational sex: There is an urgent need to discourage young women from engaging in age-disparate relationships because of the increased risk of HIV-infection involved. Targeted communication interventions are required to raise awareness and understanding of the risk of such relationships. Additionally there is a need to intensify efforts to change social norms that accept such practices. Since age-disparate relationships have been associated with financial gain, it is necessary to ensure that young females are empowered and have access to appropriate financial support (Shisana, Rehle, Simbayi et al, 2014).
- Responsible alcohol consumption: Devise innovative campaigns on responsible alcohol consumption, prioritising African male students in Gauteng and KwaZulu-Natal.
- Positive prevention: Facilitate the active involvement of HIV-positive students and staff being in prevention activities. Develop support groups of HIV-positive students and staff that emphasise the prevention of HIV transmission.

5.2.3. Structural HIV prevention interventions

The TVET sector should develop structural HIV prevention interventions that would impact positively on students and staff. These should include, but not be limited to, the items outlined below:

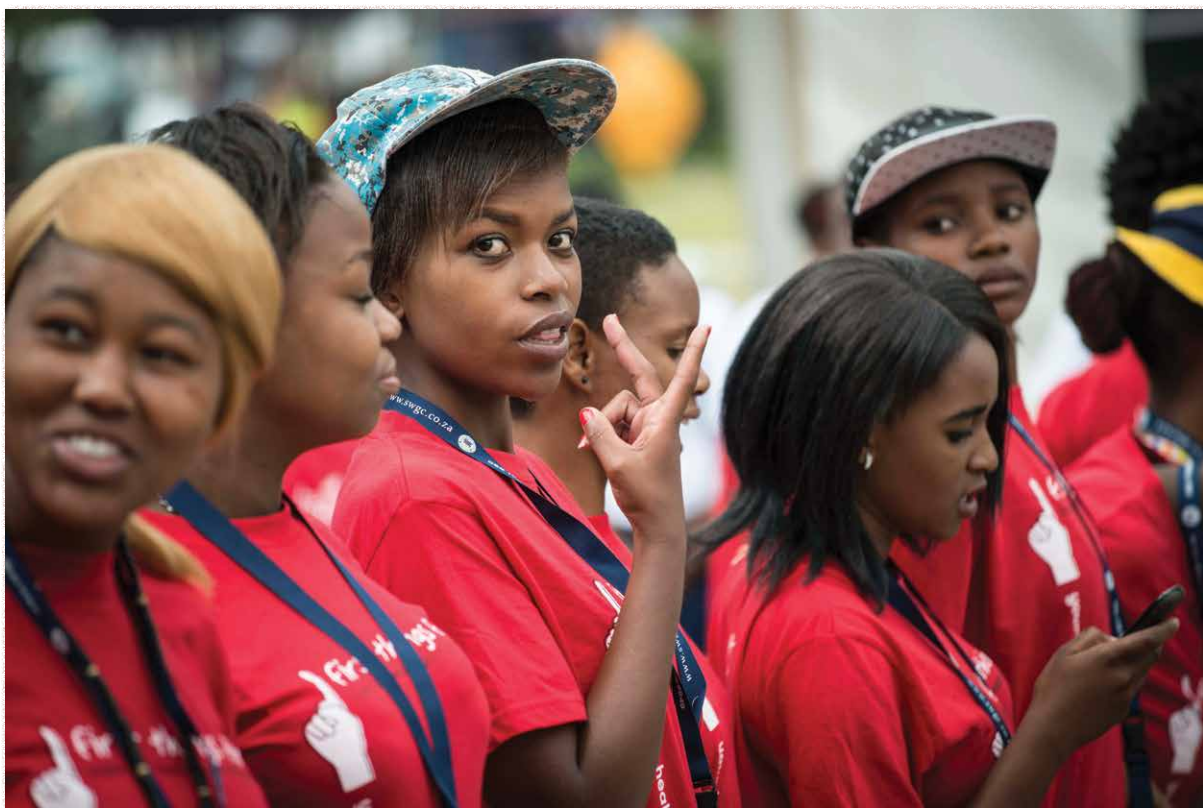
Education at a TVET college is, on its own, a structural intervention which has been shown to reduce the risk of HIV among young people and young women, in particular. In this regard, it is important to ensure that the quality of education provided at TVET colleges is enhanced to achieve high pass rates and high throughput rates so that more students benefit from further education.

Another structural HIV-prevention intervention appropriate to the TVET sector is facilitating access to the *National Student Financial Aid Scheme (NSFAS)*. Providing students with a source of financial support for their studies would reduce their risk of HIV as they would be less likely to become involved in intergenerational and transactional sexual relationships for financial gain (Cluver, Boyes, Orkin et al, 2013). This suggests a need to strengthen and extend coverage of the NSFAS scheme at TVET colleges.

The sector's response to HIV, TB and STIs needs to be put *on a firm footing* and involve student leadership, student affairs, staff unions and management structures.

5.2.4. Multi-sectoral approach to HIV programme delivery

Some TVETs have adopted a multi-sectoral approach to delivering HIV programmes and this appears to have been well received by both students and staff. In Limpopo, for example, in addition to TVET college leadership, counsellors and peer educators, the Department of Health, the Department of Social Development, non-governmental organisations and other partners participate in programme planning and implementation.



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Appendix 1: List of TVET colleges by province

	Province	College Name
1	Eastern Cape	Buffalo City TVET College
2	Eastern Cape	East Cape Midlands TVET College
3	Eastern Cape	Ikhala TVET College
4	Eastern Cape	Ingwe TVET College
5	Eastern Cape	King Hintsa TVET College
6	Eastern Cape	King Sabata Dalindyebo TVET College
7	Eastern Cape	Lovedale TVET College
8	Eastern Cape	Port Elizabeth TVET College
9	Free State	Flavius Mareka
10	Free State	Goldfields TVET College
11	Free State	Maluti TVET College
12	Free State	Motheo TVET College
13	Gauteng	Ekurhuleni East TVET College
14	Gauteng	Central JHB
15	Gauteng	Ekurhuleni West College
16	Gauteng	Sedibeng TVET College
17	Gauteng	South West TVET College
18	Gauteng	Tshwane North TVET College
19	Gauteng	Tshwane South TVET College
20	Gauteng	Western College TVET
21	KwaZulu Natal	Coastal TVET College (Mobeni)
22	KwaZulu Natal	Elangeni TVET College
23	KwaZulu Natal	Esayidi TVET College
24	KwaZulu Natal	Majuba TVET College
25	KwaZulu Natal	Mnambithi TVET College
26	KwaZulu Natal	Mthashana TVET College
27	KwaZulu Natal	Thekwini TVET College
28	KwaZulu Natal	Umfoloji TVET College
29	KwaZulu Natal	Umgungu-ndlovu TVET College
30	Limpopo	Capricorn TVET College
31	Limpopo	Lephalale TVET College
32	Limpopo	Letaba TVET College
33	Limpopo	Mopani South East TVET College

34	Limpopo	Sekhukhune TVET College
35	Limpopo	Vhembe TVET College
36	Limpopo	Waterberg TVET College
37	Mpumalanga	Ehlanzeni TVET College
38	Mpumalanga	Gert Sibande TVET College
39	Mpumalanga	Nkangala TVET College
40	Northern Cape	Northern Cape Rural TVET College
41	Northern Cape	Northern Cape Urban TVET College
42	North West	Orbit TVET College
43	North West	Taletso TVET College
44	North West	Vuselela TVET College
45	Western Cape	Boland TVET College
46	Western Cape	College of Cape Town TVET College
47	Western Cape	False Bay TVET College
48	Western Cape	Northlink TVET College
49	Western Cape	South Cape TVET College
50	Western Cape	West Coast TVET College





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